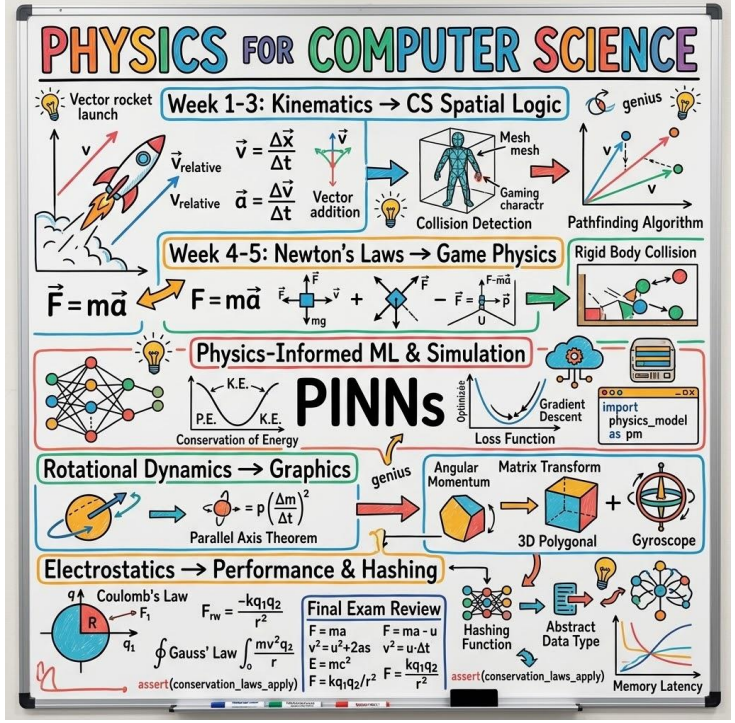
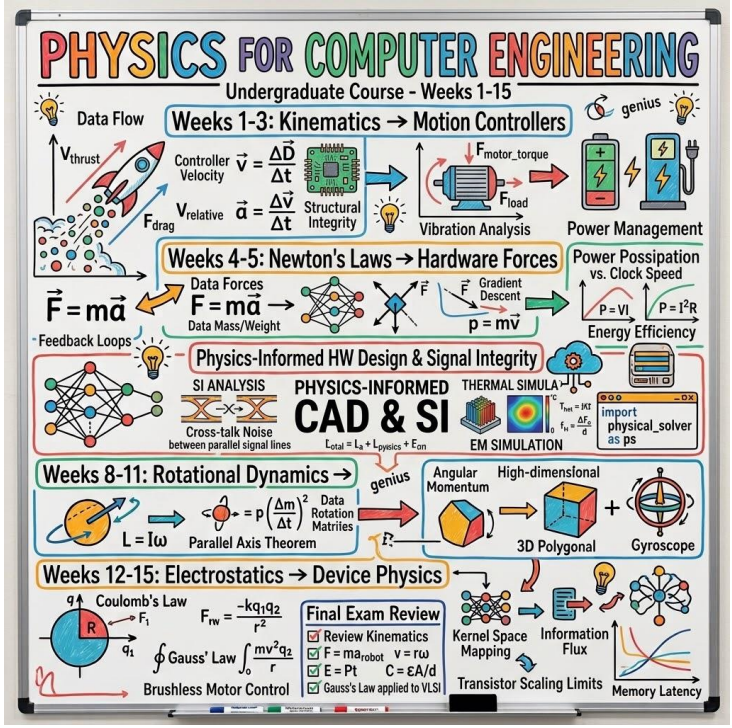
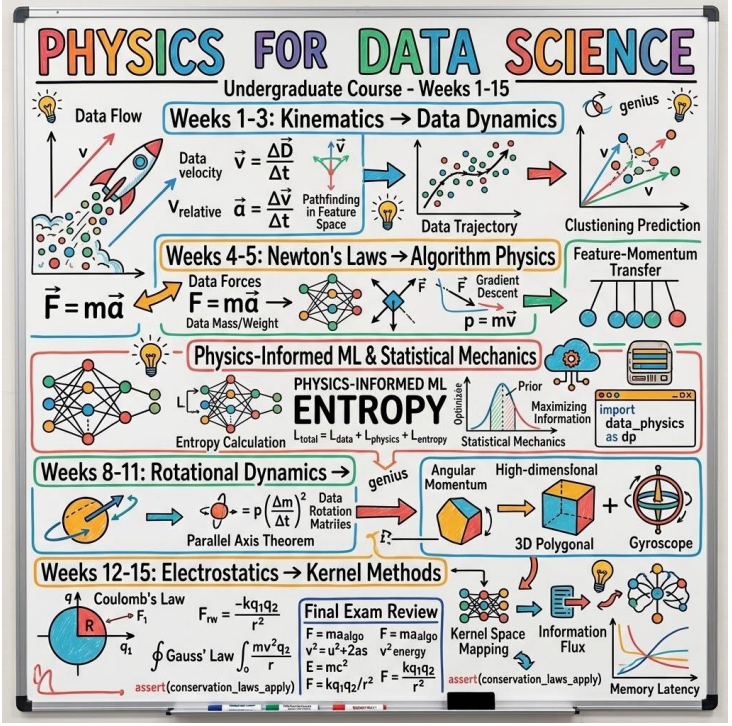
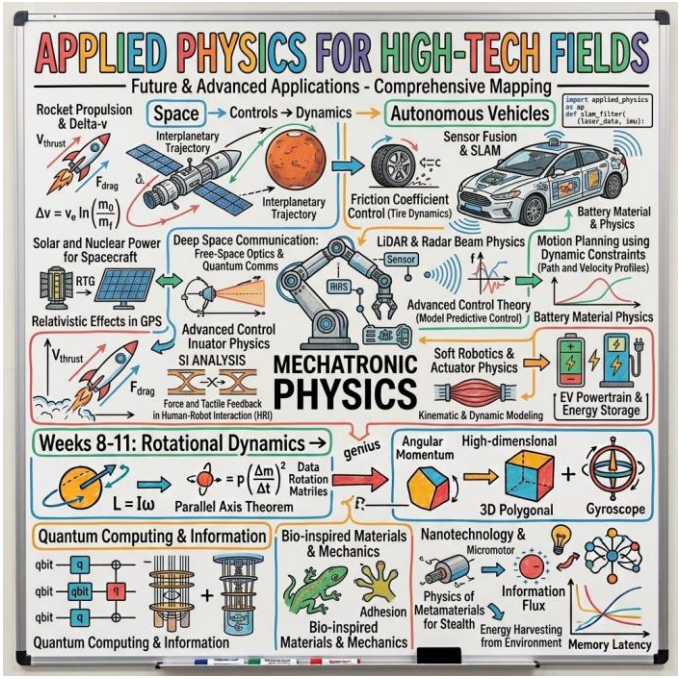








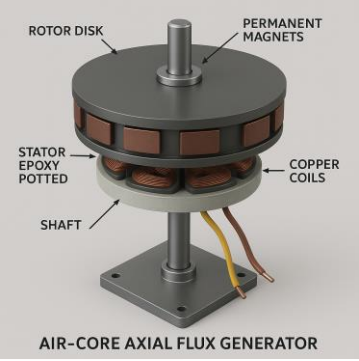
Applied Physics
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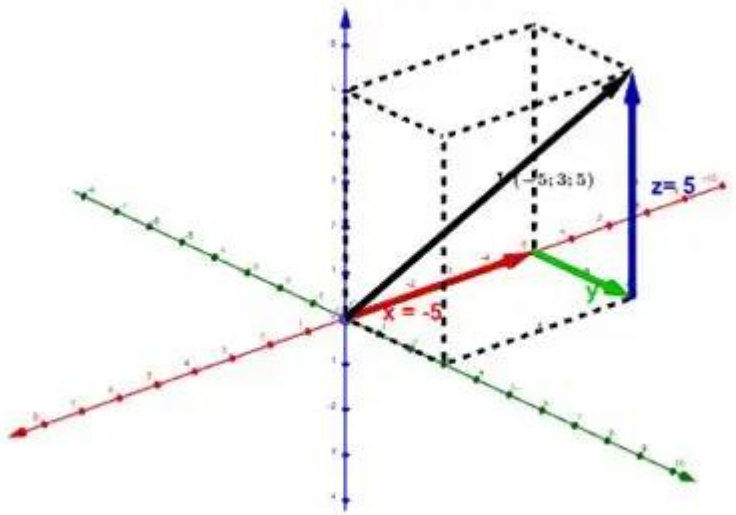
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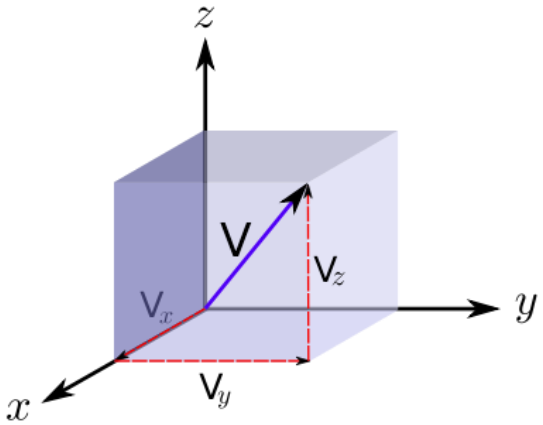
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Weekly Plan



- **Vector's Operations**
- **Numerical Examples**
- **Motion in Two or Three Dimensions**





Equations of Motion with Constant Acceleration

Projectile motion is a direct application of the 1D equations to two dimensions (x and y) independently.

The acceleration is constant: $a_x = 0$ and $a_y = -g$.

Position as a function of time:

- $x = x_0 + (v_0 \cos\theta)t$
- $y = y_0 + (v_0 \sin\theta)t - \frac{1}{2}gt^2$

Velocity as a function of time:

- $v_x = v_0 \cos\theta$ (constant)
- $v_y = v_0 \sin\theta - gt$

Projectile motion = Horizontal motion (uniform velocity) + Vertical motion (uniform acceleration)

Equations of Motion with Constant Acceleration

Projectile motion = Horizontal motion (uniform velocity) + Vertical motion (uniform acceleration)

These two motions:

- Happen simultaneously
- Do not affect each other
- Can be solved separately
- Are combined only at the end for the final answer

Gravity Acts Only Vertically

Gravity pulls the projectile downward only. It has:

No horizontal component ($g_x = 0$)

Only vertical component ($g_y = -9.8\ m/s^2$)

The Two Possible Sign Conventions

You have two choices for your 1D axis:

Convention A: Upward is Positive (+)

- g is negative: $a = -g = -9.8\ m/s^2$
- Downward motion → negative velocity
- Objects thrown up → positive initial velocity

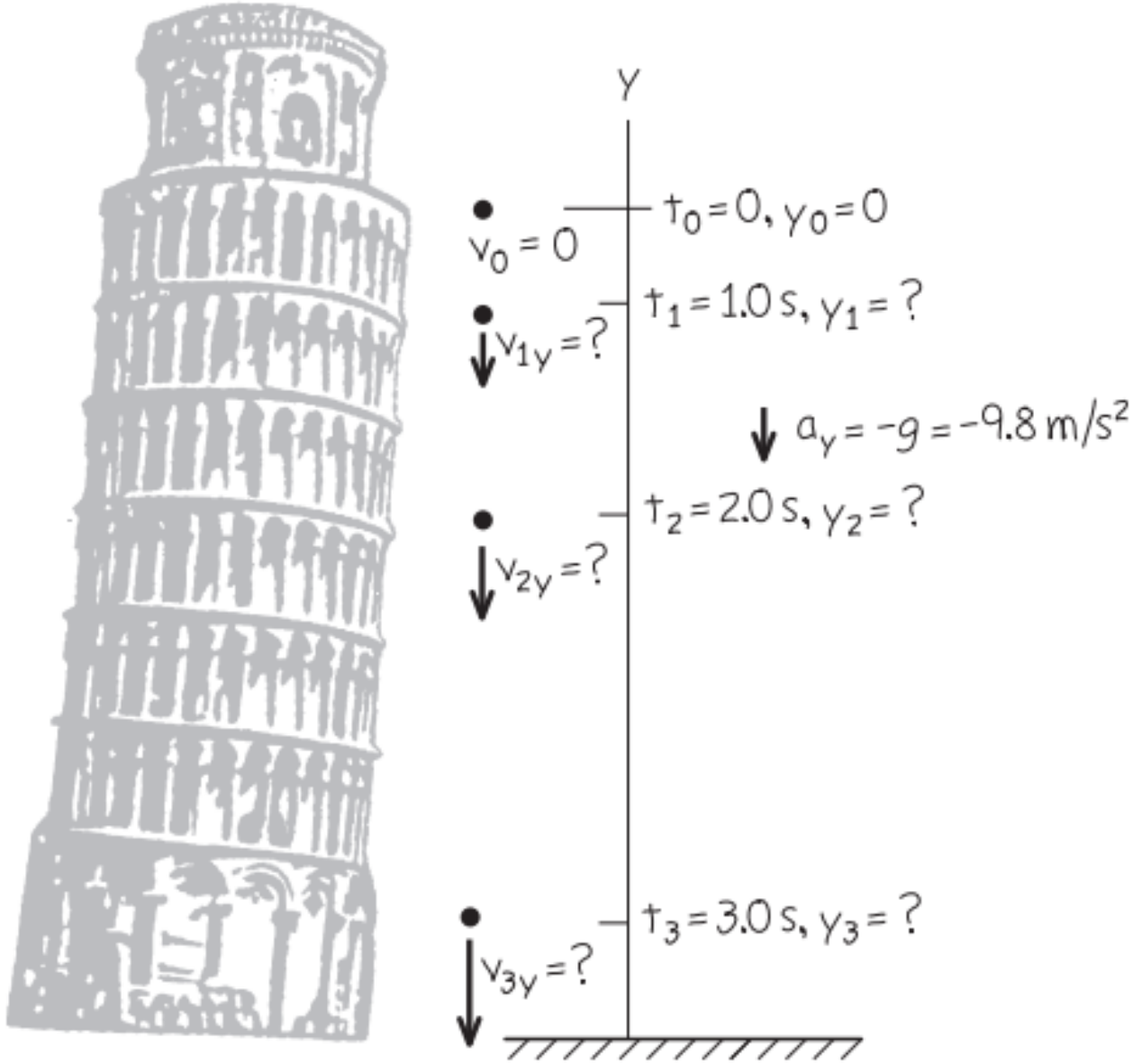
Convention B: Downward is Positive (+)

- g is positive: $a = +g = +9.8\ m/s^2$
- Downward motion → positive velocity
- Objects thrown up → negative initial velocity

Both conventions give the same physical answers.

Example

A one-euro coin is dropped from the Leaning Tower of Pisa and falls freely from rest. What are its position and velocity after 1.0 s, 2.0 s, and 3.0 s? Ignore air resistance.



Example

$$y = ?\,, t = 1,2,3\ s$$

$$v_y = ?\,, t = 1,2,3\ s$$

"Falls freely" means "falls with constant acceleration due to gravity," so we can use the constant-acceleration equations.

Vertical coordinate axis and the coordinate y instead of x.

Both the initial coordinate y_0 and initial y-velocity v_{0y} are zero.

The y-acceleration is downward (in the negative y-direction),

$$a_y = -g = -9.8\ m/s^2$$

(Remember that g is a positive quantity.)

$$y = y_0 + v_{0y}t + (1/2)a_y t^2$$

$$v_y = v_{0y} + at$$

Example

$$y = ?\,, t = 1,2,3\ s$$

$$v_y = ?\,, t = 1,2,3\ s$$

$$y = y_0 + v_{0y}t + (1/2)a_y t^2$$

$$y = 0 + (0)t + (1/2)(-g)t^2$$

$$y = (-4.9)t^2$$

$$v_y = v_{0y} + at$$

$$v_y = 0 + (-g)t$$

$$v_y = 0 + (-9.8)t$$

$$y = -4.9, -20, -40\ (m)$$

$$v_y = -9.8, -20, -29\ (m/s)$$

Velocity and Position by Integration

$$\vec{v} = \frac{d\vec{x}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\Delta v_x = a_x \Delta t$$

$$v_{2x} - v_{1x} = \int_{v_{1x}}^{v_{2x}} dv_x = \int_{t_1}^{t_2} a_x dt$$

Velocity and Position by Integration

$$\vec{v} = \frac{d\vec{x}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

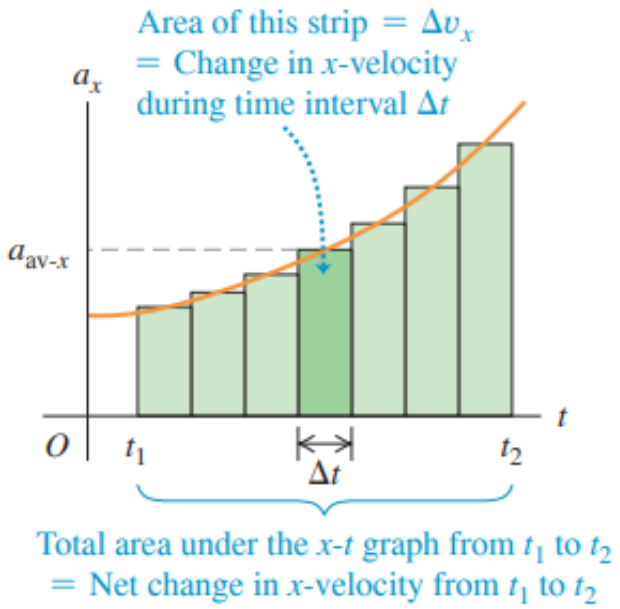
$$v_{2x} - v_{1x} = \int_{v_{1x}}^{v_{2x}} dv_x = \int_{t_1}^{t_2} a_x dt$$

$$x_2 - x_1 = \int_{x_1}^{x_2} dx = \int_{t_1}^{t_2} v_x dt$$

Velocity and Position by Integration

$$\vec{v} = \frac{d\vec{x}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$



Example

A person is driving along a straight highway. At $t = 0$, when he is moving at $10\ m/s$ in the positive $x - direction$, he passes a signpost at $x = 50\ m$. His $x - acceleration$ as a function of time is

$$a_x = 2.0\ m/s^2 - (0.1\ m/s^3)t$$

(a) Find his $x - velocity\ v_x$ and position x as functions of time.
(b) When is his $x - velocity$ greatest?
(c) What is that maximum $x - velocity$?
(d) Where is the car when it reaches that maximum $x - velocity$?

(a) Find his $x-velocity\ v_x$ and position x as functions of time.

Example

At $t = 0$, when he is moving at $10\ m/s$ in the positive $x-direction$, he passes a signpost at $x = 50\ m$

$$a_x = 2.0\ m/s^2 - (0.1\ m/s^3)t$$

(a)

Velocity:

$$v_x(t) = v_x(0) + \int_0^t a_x(t')\,dt'$$

$$= 10 + \int_0^t (2.0 - 0.1t')\,dt'$$

$$= 10 + [2.0t' - 0.05t'^2]_0^t$$

$$= 10 + 2.0t - 0.05t^2$$

So:

$$\boxed{v_x(t) = 10 + 2.0t - 0.05t^2 \text{ m/s}}$$

Position:

$$x(t) = x(0) + \int_0^t v_x(t')\,dt'$$

$$= 50 + \int_0^t (10 + 2.0t' - 0.05t'^2)\,dt'$$

$$= 50 + \left[10t' + 1.0t'^2 - \frac{0.05}{3}t'^3\right]_0^t$$

$$= 50 + 10t + 1.0t^2 - \frac{0.05}{3}t^3$$

Note: $\frac{0.05}{3} = \frac{1}{60} \approx 0.016667.$

$$\boxed{x(t) = 50 + 10t + t^2 - \frac{1}{60}t^3 \text{ m}}$$

Example

(b) When is his $x-velocity$ greatest?
(c) What is that maximum $x-velocity$?

(b)

When is v_x greatest?

$v_x(t)$ **is greatest when** $a_x(t) = 0$ **(since** $a_x = dv_x/dt$**).**

$$2.0 - 0.1t = 0$$

$$t = \frac{2.0}{0.1} = 20 \text{ s}$$

(c)

Maximum v_x

$$v_x(20) = 10 + 2(20) - 0.05(20)^2$$

$$= 10 + 40 - 0.05(400)$$

$$= 50 - 20 = 30 \text{ m/s}$$

(d) Where is the car when it reaches that maximum $x - velocity$?

Example

(d)

Position at maximum velocity

$$x(20) = 50 + 10(20) + (20)^2 - \frac{(20)^3}{60}$$

$$= 50 + 200 + 400 - \frac{8000}{60}$$

$$= 650 - 133.333 \ldots$$

$$= 516.666 \ldots \textbf{ m}$$

Example

During your summer internship for an aerospace company, you are asked to design a small research rocket. The rocket is to be launched from rest from the earth's surface and is to reach a maximum height of $960\ m$ above the earth's surface. The rocket's engines give the rocket an upward acceleration of $16.0\ m/s^2$ during the time T that they fire. After the engines shut off, the rocket is in free fall. Ignore air resistance.

What must be the value of T in order for the rocket to reach the required altitude?

Example

The rocket accelerates uniformly upward at $16.0\ m/s^2$ with the engines on. After the engines are off, it moves upward but accelerates downward at $9.80\ m/s^2$.

$$y = y_0 + v_{0y}t + (1/2)a_y t^2$$

$$v_y{}^2 = v_{0y}{}^2 + 2a_y(y - y_0)$$

both apply to both parts of the motion since the accelerations are both constant, but the accelerations are different in both cases. Let $+y$ be upward.

Example

With the engines on, $v_{oy} = 0$, $a_y = 16.0\ m/s^2$ upward, and $t = T$ at the instant the engines just shut off. Using these quantities,

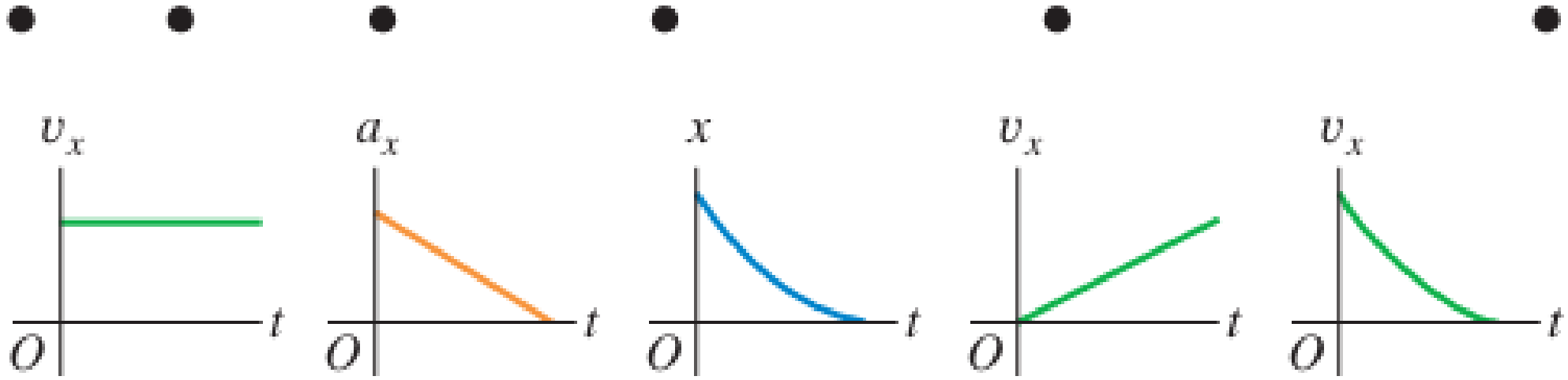
$$y = y_0 + v_{0y}t + (1/2)a_x t^2 = (1/2)(16)T^2$$

$$v_y{}^2 = v_{0y}{}^2 + 2ay(y - y_0) = (16)T$$

Technique	Model/Method	Suitable for
1. Analytical model	$x = x_0 + v_0 t + \frac{1}{2} a t^2$	Basic mechanics
2. Graphical model	$x - t,\ v - t,\ a - t$graphs	Teaching/visualization
3. Numerical modeling	Euler, Verlet, RK4	Computer simulation
4. Sensor-based model	Ultrasonic/IR/LiDAR sensor	Real-time measurement
5. Video analysis	Track an object's position frame-by-frame	Experimental physics
6. Encoder-based model	Rotary encoder → linear displacement	Robotics/mechatronics
7. Computer simulation	Python/MATLAB/Simulink	Simulation and analysis
8. AI/ML model	Measured t→ predicted x	Advanced modeling
9. Digital twin	Physical object + mathematical/computational model	Advanced laboratory
10. Interactive model	Slider/time input → moving object + graphs	Physics teaching

Example

The black dots at the top of figure represent a series of high speed photographs of an insect flying in a straight line from left to right (in the positive $x - direction$). Which of the graphs in figure most plausibly depicts this insect's motion?



Equations of Motion with Constant Acceleration

$$v_x = v_{0x} + at$$

$$x = x_0 + v_{0x}t + (1/2)a_x t^2$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

$$x - x_0 = (1/2)(v_x + v_{0x})t$$

TABLE 2.5 Equations of Motion with Constant Acceleration

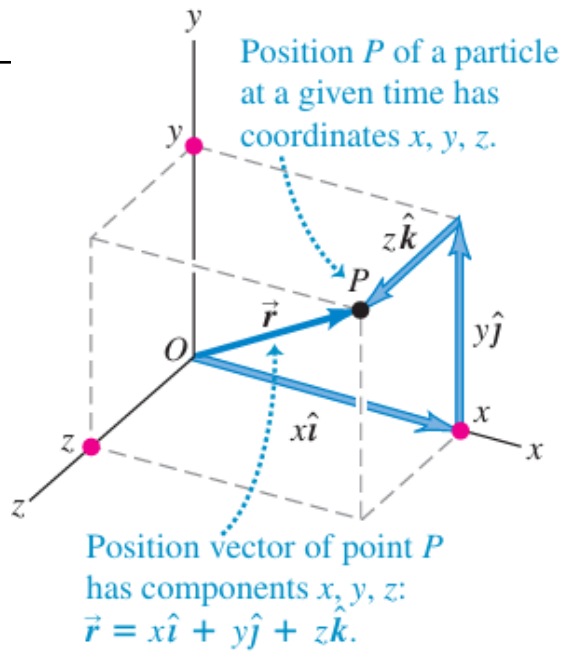
Equation		Includes Quantities			
$v_x = v_{0x} + a_x t$	(2.8)	t		v_x	a_x
$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$	(2.12)	t	x		a_x
$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	(2.13)		x	v_x	a_x
$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$	(2.14)	t	x	v_x	

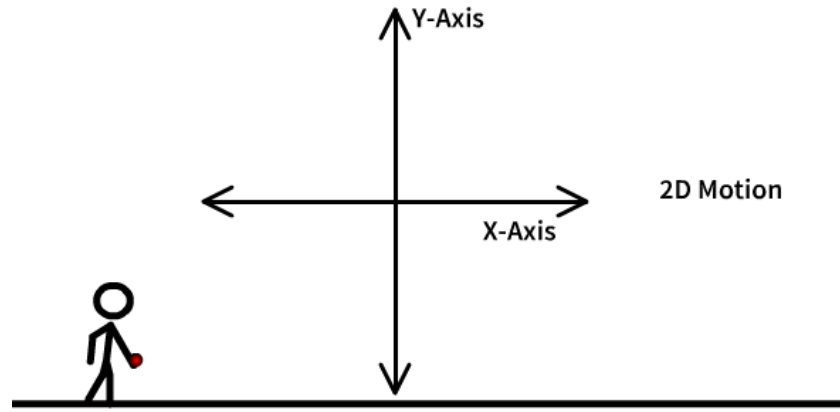
Position

A particle is at a point P at a certain instant, the position vector $\vec{r}$ of the particle at this instant is a vector that goes from the origin of the coordinate system to point P

The Cartesian coordinates x, y, and z of point P are the x-, y-, and z-components of vector $\vec{r}$.

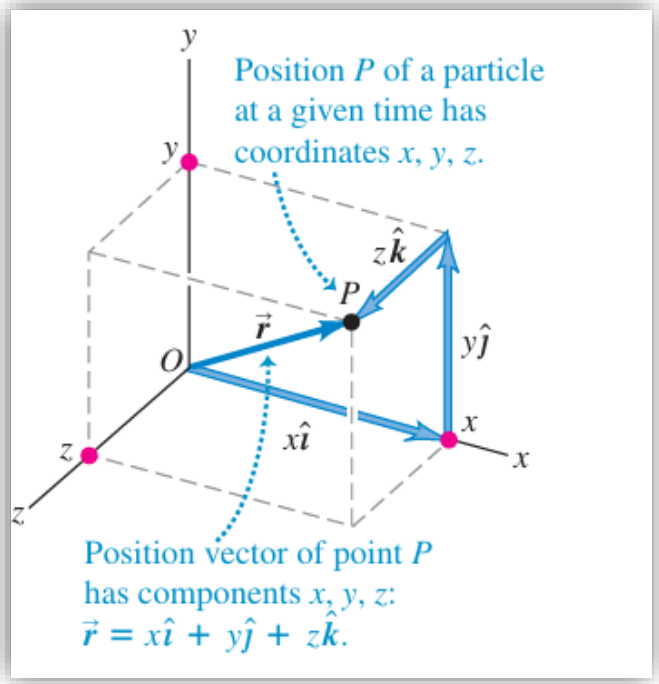
$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$



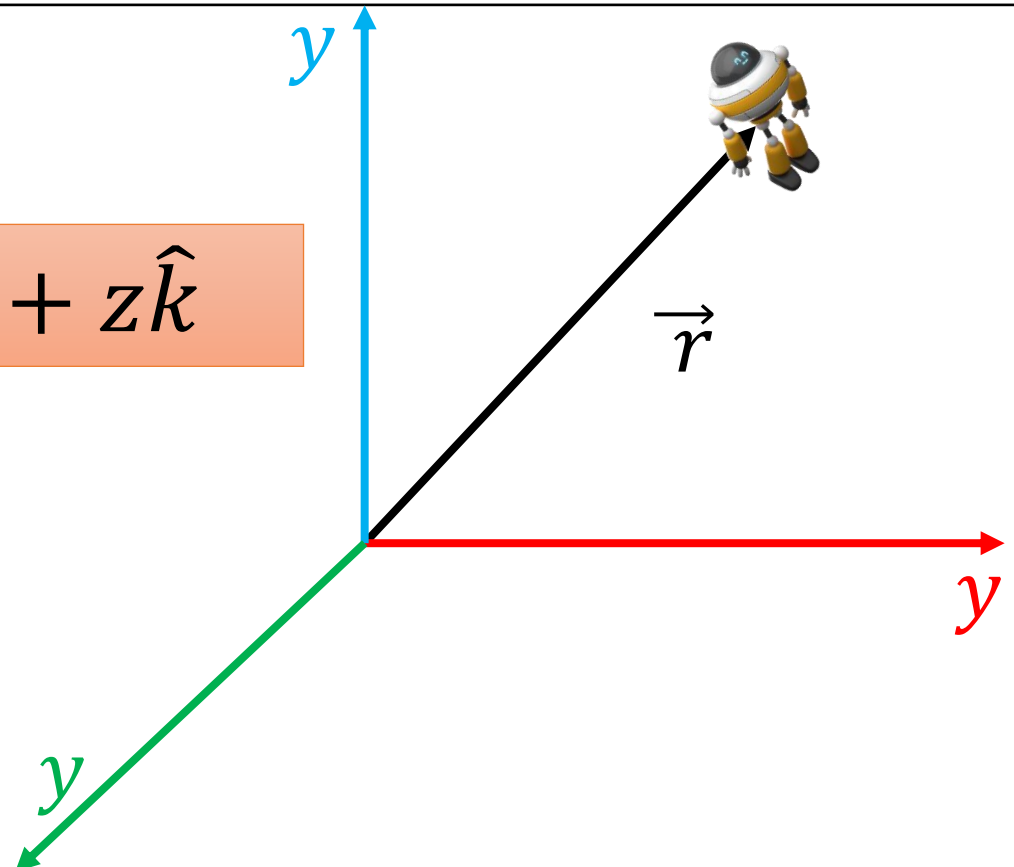


Displacement

A ***position vector*** **r** is defined as a fixed vector which locates a point in space relative to another point.



$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$

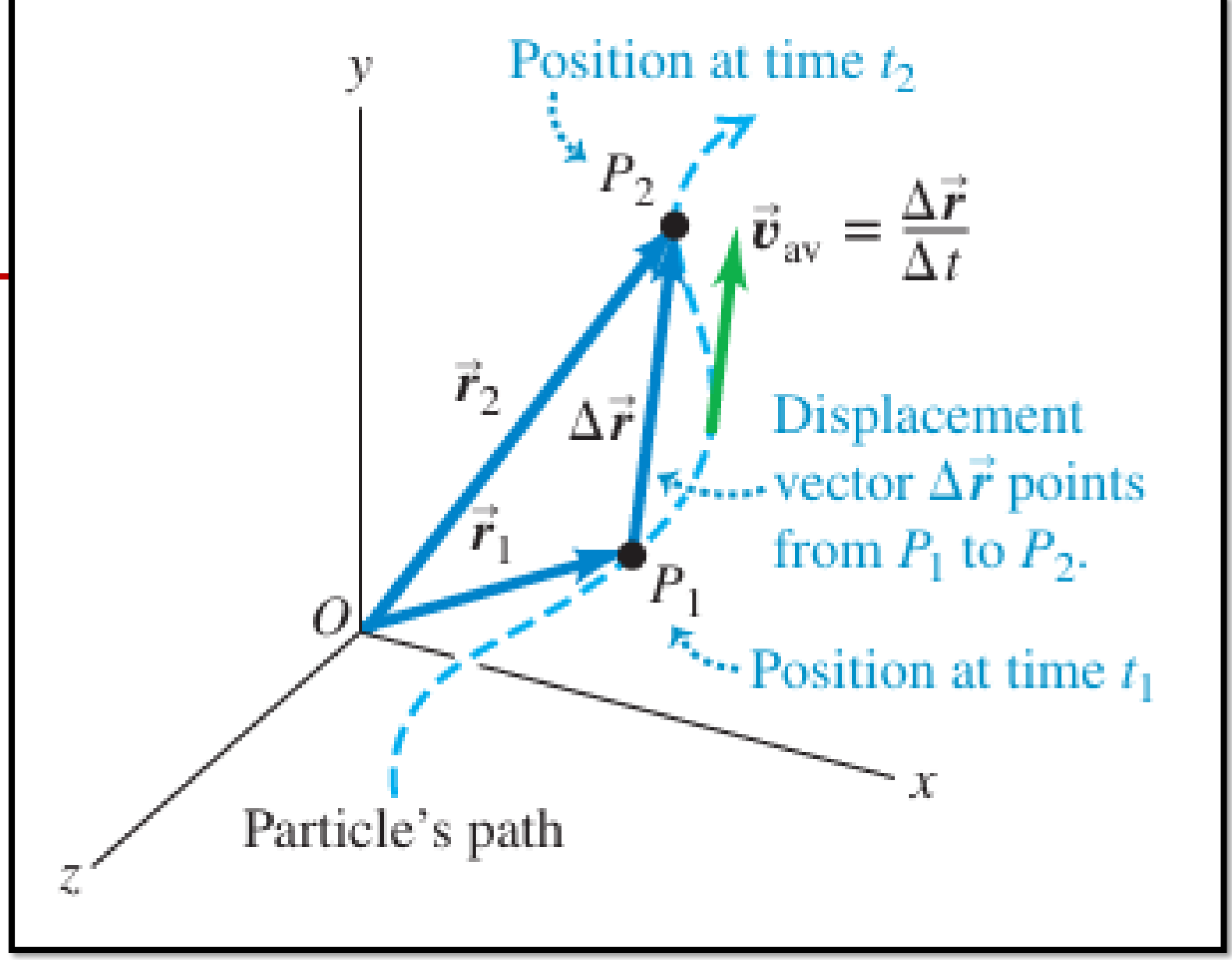


Displacement

$$\overrightarrow{r_1} = x_1\hat{\imath} + y_1\hat{\jmath} + z_1\hat{k}$$

$$\overrightarrow{r_2} = x_2\hat{\imath} + y_2\hat{\jmath} + z_2\hat{k}$$

$$\overrightarrow{r_2} - \overrightarrow{r_2} = \overrightarrow{\Delta r}$$



$$\Delta r = \overrightarrow{r_2} - \overrightarrow{r_1} = (x_2\hat{\imath} + y_2\hat{\jmath} + z_2\hat{k}) - (x_1\hat{\imath} + y_1\hat{\jmath} + z_1\hat{k})$$

Displacement, Velocity



$$\Delta \boldsymbol{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = (x_2\hat{\imath} + y_2\hat{\jmath} + z_2\hat{k}) - (x_1\hat{\imath} + y_1\hat{\jmath} + z_1\hat{k})$$

$$\overrightarrow{V_{av}} = \frac{\Delta \boldsymbol{r}}{\Delta t} = \frac{\overrightarrow{r_2} - \overrightarrow{r_1}}{t_2 - t_1}$$

$$\overrightarrow{V_{ins}} = \lim_{n \to 0} \frac{\Delta \boldsymbol{r}}{\Delta t}$$

$$\overrightarrow{V_{ins}} = \frac{d\boldsymbol{r}}{dt} = \frac{d}{dt}(x\hat{\imath} + y\hat{\jmath} + z\hat{k}) = v_x\hat{\imath} + v_y\hat{\jmath} + v_z\hat{k}$$

Example



A robotic vehicle, or rover, is exploring the surface of Mars. The stationary Mars lander is the origin of coordinates, and the surrounding Martian surface lies in the xy-plane. The rover, which we represent as a point, has x- and y-coordinates that vary with time:

$$x = 2.0\ m - (0.25\ m/s^2)(t)^2$$

$$y = (1.0\ m/s)t + (0.025\ m/s^3)(t)^3$$

(a) Find the rover's coordinates and distance from the lander at $t = 2.0\ s$.

(b) Find the rover's displacement and average velocity vectors for the interval $t = 0.0\ s$ to $t = 2.0\ s$.

(c) Find a general expression for the rover's instantaneous velocity vector $\vec{V}$.

(d) Express $\vec{V}$ at $t = 2.0$ s in component form and in terms of magnitude and direction.

Example

Find the rover's coordinates and distance from the lander at $t = 2.0\ s$.

(a)

$$x = 2.0 - (0.25)(t)^2$$

$$x = 2.0 - (0.25)(2)^2 = 1.0\ m$$

$$y = (1.0)t + (0.025)(t)^3$$

$$y = (1.0)(2) + (0.025)(2)^3 = 2.2\ m$$

Distance from lander (origin):

$$r = \sqrt{x^2 + y^2} = \sqrt{(1.0)^2 + (2.2)^2} = \sqrt{1.0 + 4.84} = \sqrt{5.84}$$

$$r \approx 2.4166 \text{ m}$$

$$\boxed{r \approx 2.42 \text{ m}}$$

Example

Find the rover's displacement and average velocity vectors for the interval t =0.0 s to t =2.0 s.

(b)

$$x = 2.0 - (0.25)(t)^2$$

$$x = 2.0 - (0.25)(0)^2 = 2.0 \text{ m}$$

$$x = 2.0 - (0.25)(2)^2 = 1.0 \text{ m}$$

$$\overrightarrow{r_0} = 2\,\hat{\imath} + 0\hat{\jmath}$$

$$y = (1.0\,)t + (0.025\,)(t)^3$$

$$\overrightarrow{r_2} = 1\hat{\imath} + 2.2\,\hat{\jmath}$$

$$y = (1.0\,)\,(0) + (0.025\,)(0)^3 = 0 \text{ m}$$

$$y = (1.0\,)\,(2) + (0.025\,)(2)^3 = 2.2 \text{ m}$$

Example

$$\overrightarrow{r_0} = 2\,\hat{\imath} + 0\hat{\jmath}$$

$$\overrightarrow{r_2} = 1\hat{\imath} + 2.2\,\hat{\jmath}$$

$$\Delta\boldsymbol{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = (x_2\hat{\imath} + y_2\hat{\jmath} + z_2\hat{k}) - (x_1\hat{\imath} + y_1\hat{\jmath} + z_1\hat{k})$$

$$\Delta\boldsymbol{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = (x_2\hat{\imath} + y_2\hat{\jmath}) - (x_1\hat{\imath} + y_1\hat{\jmath})$$

$$\Delta\boldsymbol{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = (1\hat{\imath} + 2.2\hat{\jmath}) - (2\hat{\imath} + 0\hat{\jmath})$$

$$\Delta\boldsymbol{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = -1\hat{\imath} + 2.2\hat{\jmath}$$

$$\overrightarrow{V_{av}} = \frac{\Delta\boldsymbol{r}}{\Delta t} = \frac{\overrightarrow{r_2} - \overrightarrow{r_1}}{t_2 - t_1}$$

Example

$$\Delta \boldsymbol{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = (1\hat{\imath} + 2.2\hat{\jmath}) - (2\hat{\imath} + 0\hat{\jmath})$$

$$\Delta \boldsymbol{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = -1\hat{\imath} + 2.2\hat{\jmath}$$

$$\overrightarrow{V_{av}} = \frac{\Delta \boldsymbol{r}}{\Delta t} = \frac{-1\hat{\imath} + 2.2\hat{\jmath}}{2 - 0}$$

$$\overrightarrow{V_{av}} = \frac{\Delta \boldsymbol{r}}{\Delta t} = -0.5\hat{\imath} + 1.1\hat{\jmath}$$

Example

Find a general expression for the rover's instantaneous velocity vector $\vec{V}$.

(c)

$$x = \frac{dx}{dt} = 2.0\ m - (0.25\ m/s^2)(t)^2$$

$$V_x = 0 - 2\left(0.25\frac{m}{s^2}\right)t$$

$$y = (1.0\ m/s)t + (0.025\ m/s^3)(t)^3$$

$$V_y = \frac{dy}{dt} = (1.0\ m/s)(1) + 2(0.025\ m/s^3)(t)^2$$

Example

Find a general expression for the rover's instantaneous velocity vector $\vec{V}$.

(c)

$$V_x = 0 - 2(0.25)t$$

$$V_y = \frac{dy}{dt} = (1.0\,)(1) + 2(0.025\,)(t)^2$$

$$\overrightarrow{V} = V_x\hat{\imath} + V_y\hat{\jmath}$$

$$\overrightarrow{V} = -2(0.25)t\ \ \hat{\imath} \mathrel{+}= (1.0\,)(1) + 2(0.025\,)(t)^2\hat{\jmath}$$

Example

Express $\vec{V}$ at t =2.0 s in component form and in terms of magnitude and direction.

(c)

$$V_x = 0 - 2(0.25)(2) = -1\ m/s$$

$$V_y = (1.0)(1) + 2(0.025)(2)^2 = 1.3\ m/s$$

$$\overrightarrow{V} = -1\hat{\imath} + 1.3\hat{\jmath}$$

$$V = \sqrt{(-1)^2 + (1 \cdot 3)^2} = 1.6 m/s$$

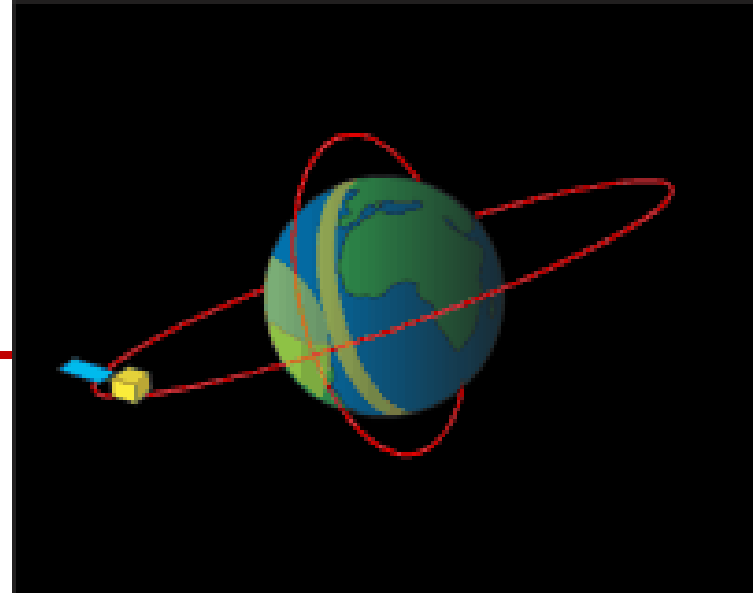
$$\emptyset = \tan(1.3/1) = 52^o$$

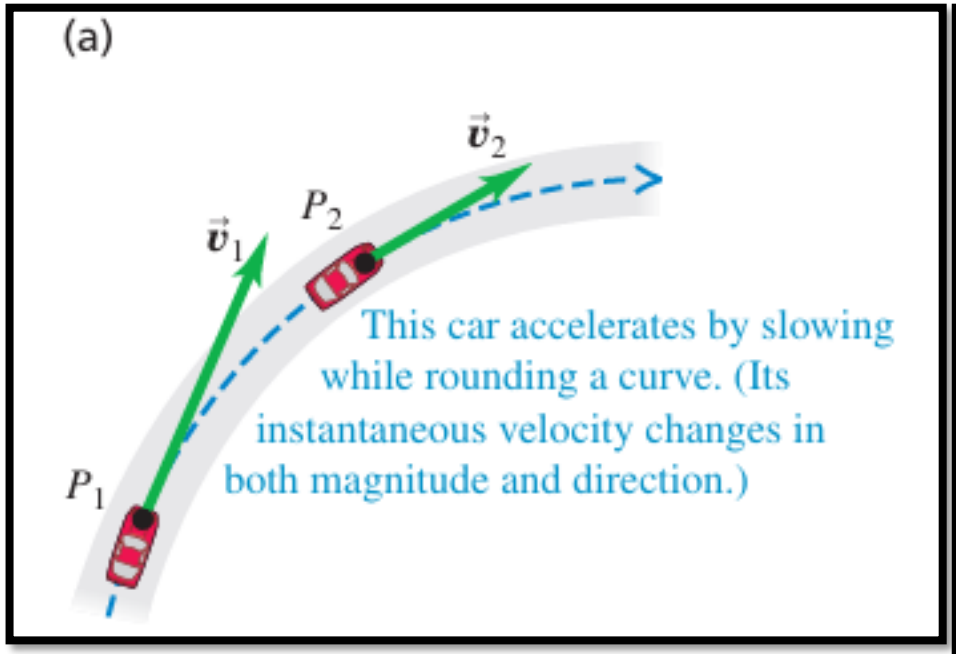
angle a between 90° and 180° with respect to the positive x-axis.

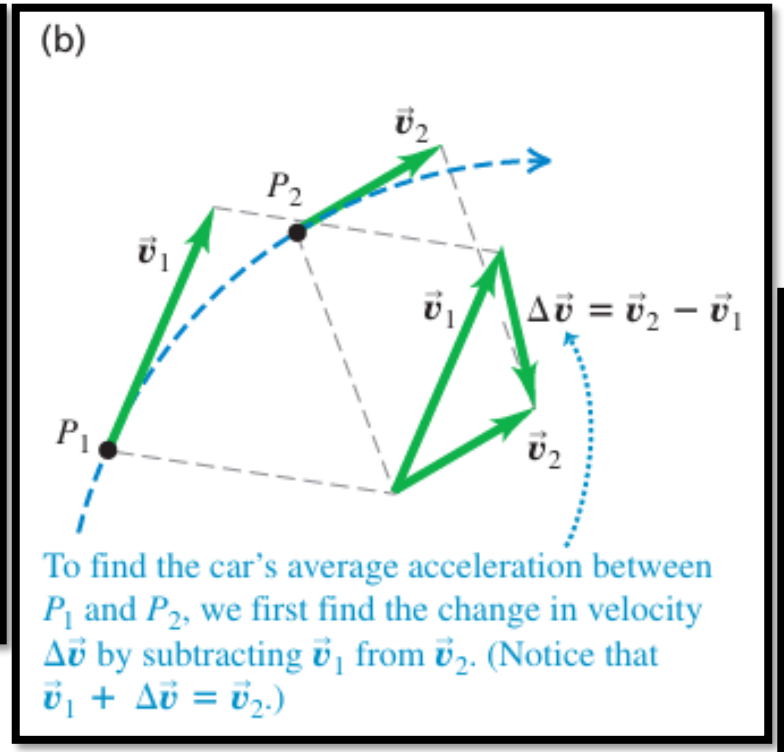
$$\emptyset = 180^o - 52^o = 128^o$$

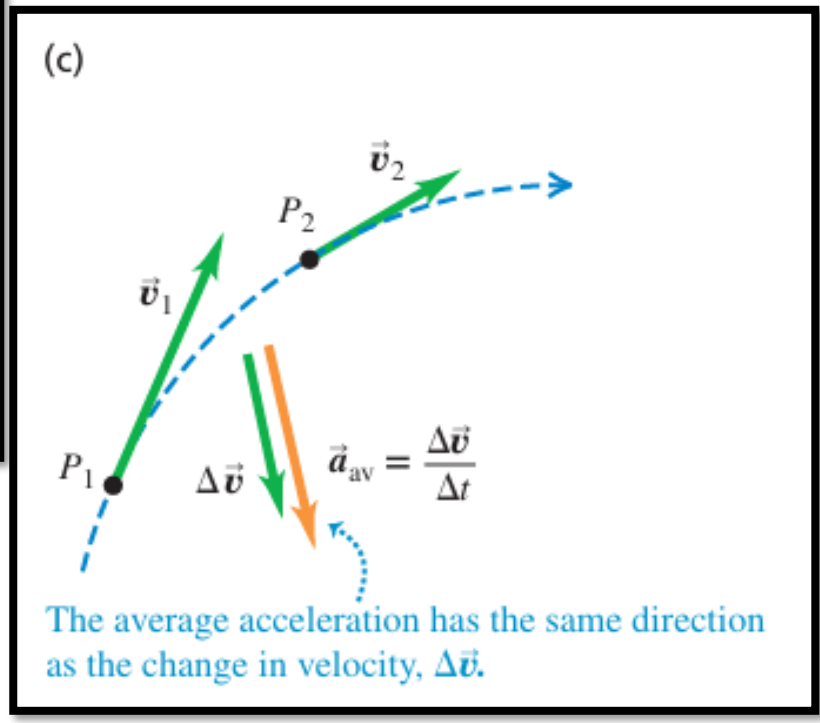
Acceleration

acceleration of a particle moving in space

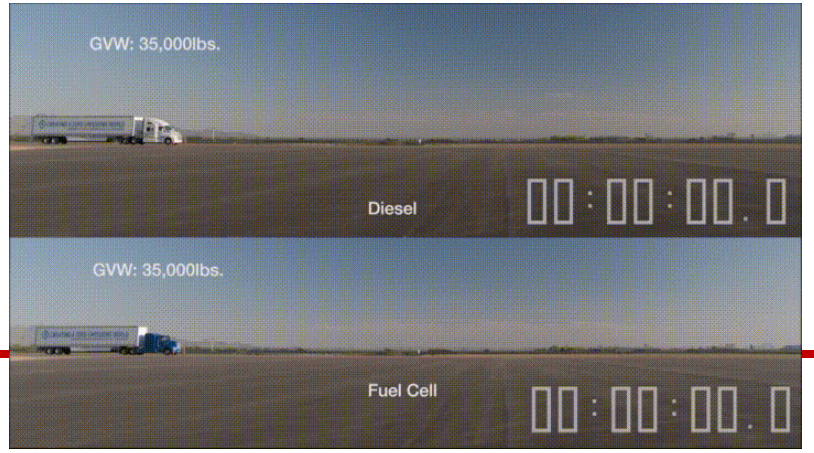








Acceleration





$$\Delta \boldsymbol{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = (x_2\hat{\imath} + y_2\hat{\jmath} + z_2\hat{k}) - (x_1\hat{\imath} + y_1\hat{\jmath} + z_1\hat{k})$$

$$\overrightarrow{V_{ins}} = \frac{d\boldsymbol{r}}{dt} = \frac{d}{dt}(x\hat{\imath} + y\hat{\jmath} + z\hat{k}) = V_x\hat{\imath} + V_y\hat{\jmath} + V_z\hat{k}$$

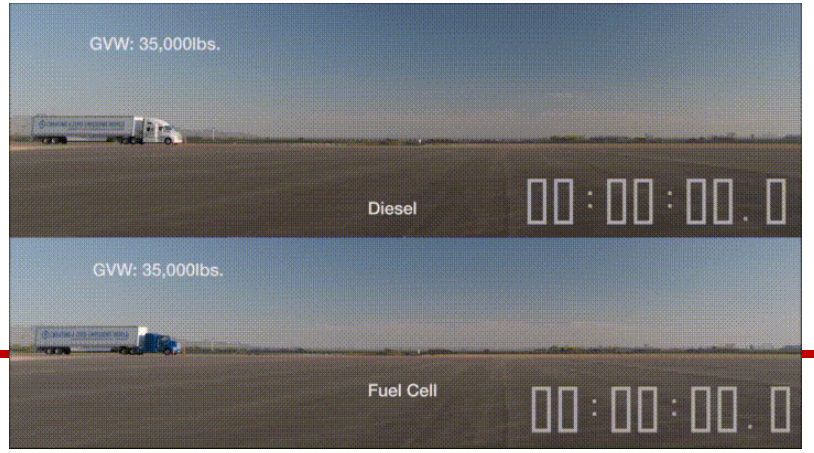
$$\overrightarrow{V_{av}} = \frac{\Delta \boldsymbol{r}}{\Delta t} = \frac{\overrightarrow{r_2} - \overrightarrow{r_1}}{t_2 - t_1}$$

$$\overrightarrow{V_{ins}} = \lim_{n \to 0} \frac{\Delta \boldsymbol{r}}{\Delta t} = \frac{d\boldsymbol{r}}{dt}$$

$$\overrightarrow{a_{av}} = \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{\overrightarrow{v_2} - \overrightarrow{v_1}}{t_2 - t_1}$$

$$\overrightarrow{a_{ins}} = \lim_{n \to 0} \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{d\boldsymbol{v}}{dt}$$

Acceleration





$$\overrightarrow{a_{av}} = \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{\overrightarrow{v_2} - \overrightarrow{v_1}}{t_2 - t_1}$$

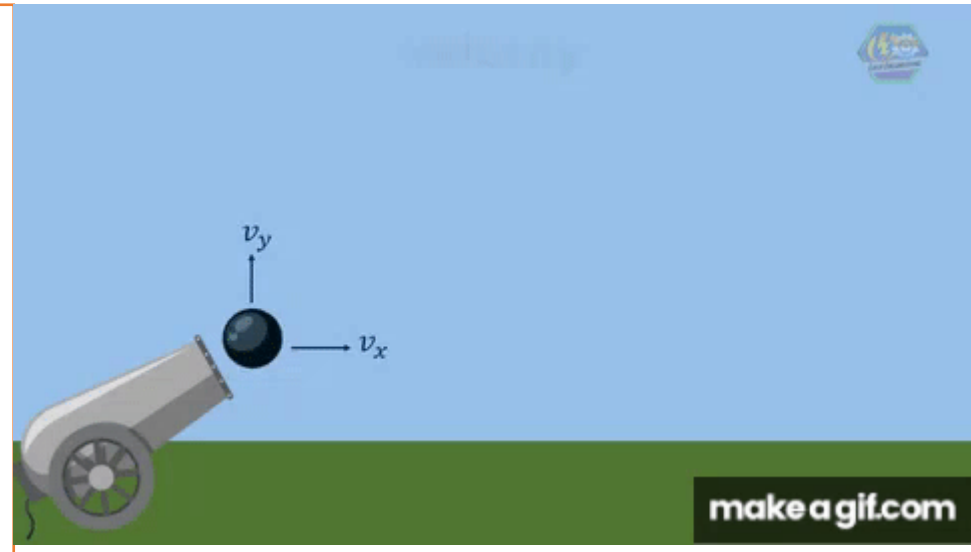
$$\overrightarrow{a_{ins}} = \lim_{n \to 0} \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{d\boldsymbol{v}}{dt}$$

$$\overrightarrow{a_{ins}} = \frac{d\boldsymbol{v}}{dt} = \frac{d}{dt}(v_x\hat{\imath} + v_y\hat{\jmath} + v_z\hat{k}) = a_x\hat{\imath} + a_y\hat{\jmath} + a_z\hat{k}$$

Projectile Motion

Projectile motion is the motion of an object thrown or projected into the air, subject only to acceleration due to gravity. The path followed by a projectile is called its trajectory, which is always a parabola (in the absence of air resistance).

- ❑ Only force acting is **gravity** (neglecting air resistance)
- ❑ Horizontal and vertical motions are **independent**
- ❑ Horizontal velocity remains **constant**
- ❑ Vertical velocity changes due to **gravity**
- ❑ Path is **parabolic**

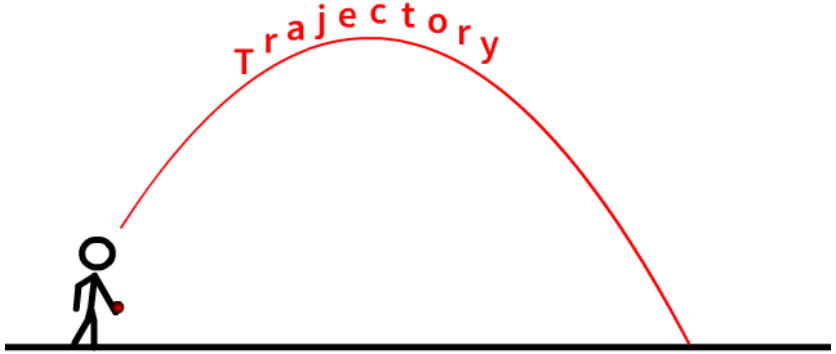


Projectile Motion

Horizontal Projection
- **Launch angle:** 0° (parallel to ground)
- **Initial vertical velocity:** 0 m/s
- **Examples:** Ball rolling off table, airplane dropping package
- **Trajectory:** Half-parabola

Oblique Projection
- **Launch angle:** Between 0° and 90°
- Has both horizontal and vertical velocity components
- **Examples:** Thrown ball, cannon firing, basketball shot
- **Trajectory:** Symmetric parabola (if launch and landing at same height)

Vertical Projection
- **Launch angle:** 90° (straight up/down)
- **Initial horizontal velocity:** 0 m/s
- **Examples:** Object thrown vertically upward
- **Trajectory:** Straight line up and down



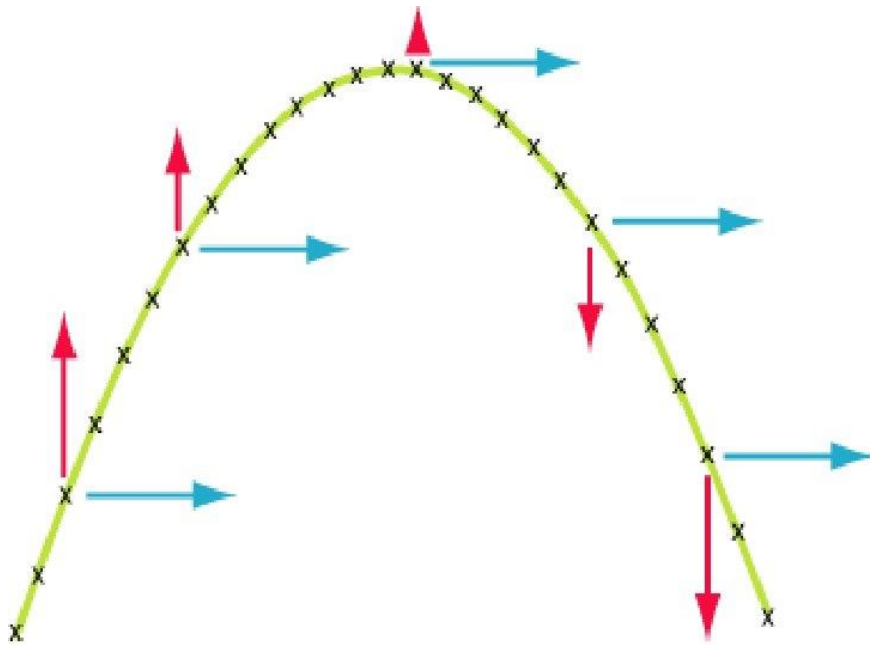
Equations of Projectile Motion

Horizontal Motion (Constant Velocity):

- **Position:** $x = x_o + v_o Cos(\theta) t$
- **Velocity:** $v_x = v_o + v_o Cos(\theta)$

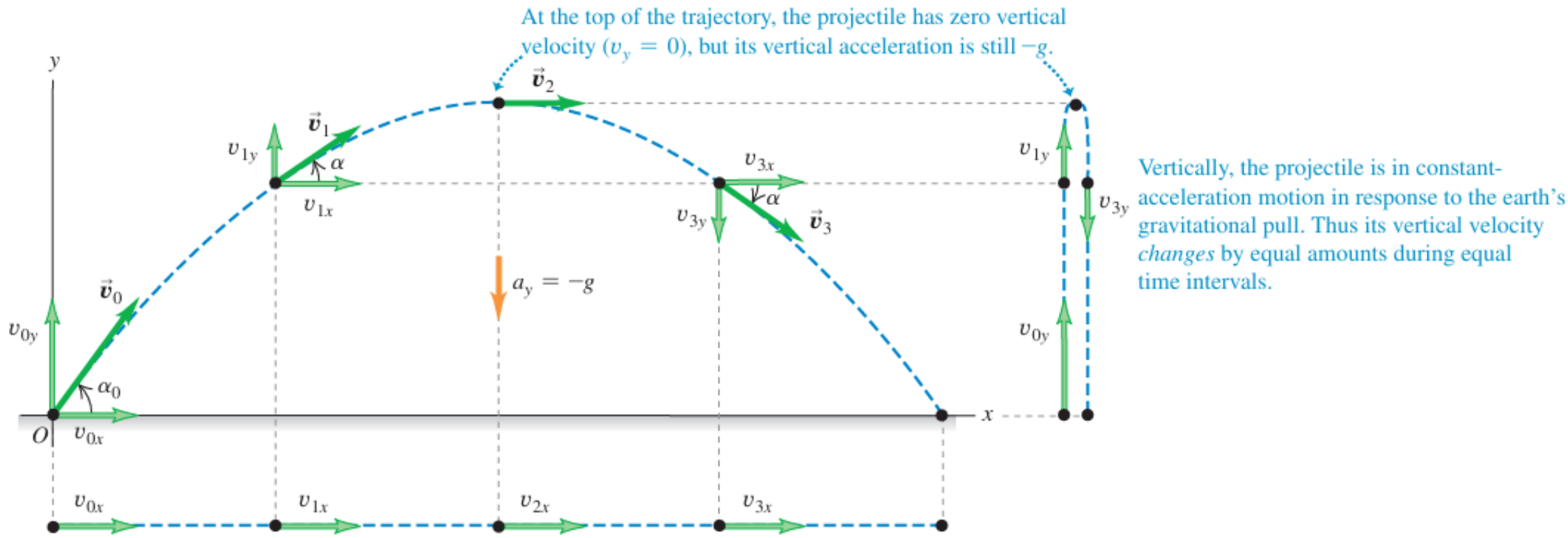
Vertical Motion (Constant Acceleration):

- **Position:** $y = y_o + v_o Sin(\theta) t - (1/2)\, g t^2$
- **Velocity:** $v_y = v_o Sin(\theta) - gt$



Equations of Projectile Motion

Figure 3.17 If air resistance is negligible, the trajectory of a projectile is a combination of horizontal motion with constant velocity and vertical motion with constant acceleration.



Equations of Projectile Motion

Time of Flight:

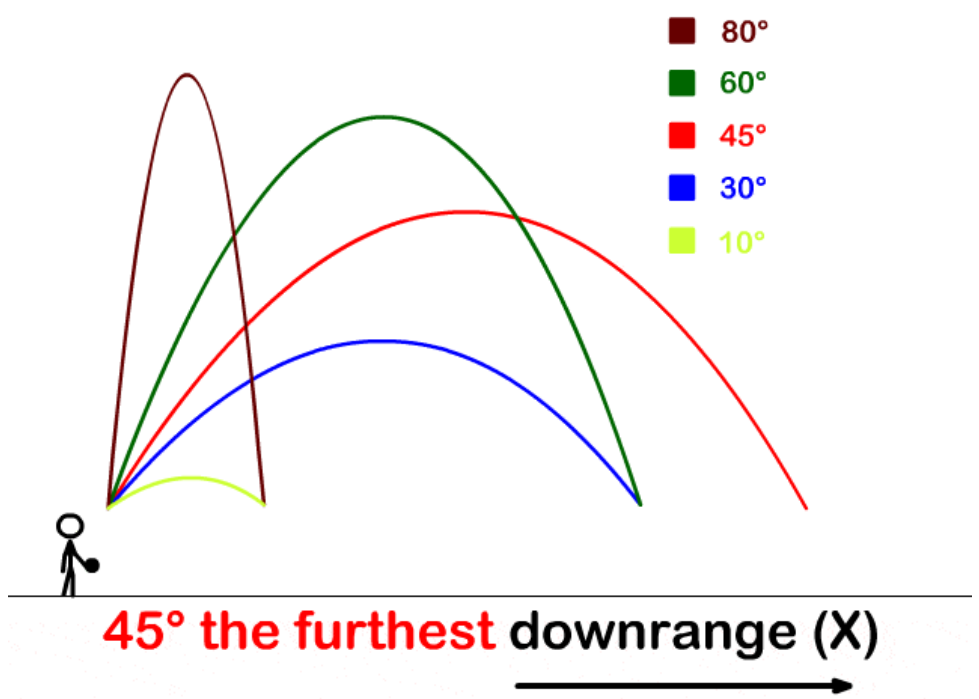
$$T = \frac{2v_o Sin(\theta)}{g}$$

Max. Height:

$$H = \frac{2v^2{}_o Sin^2(\theta)}{2g}$$

Range:

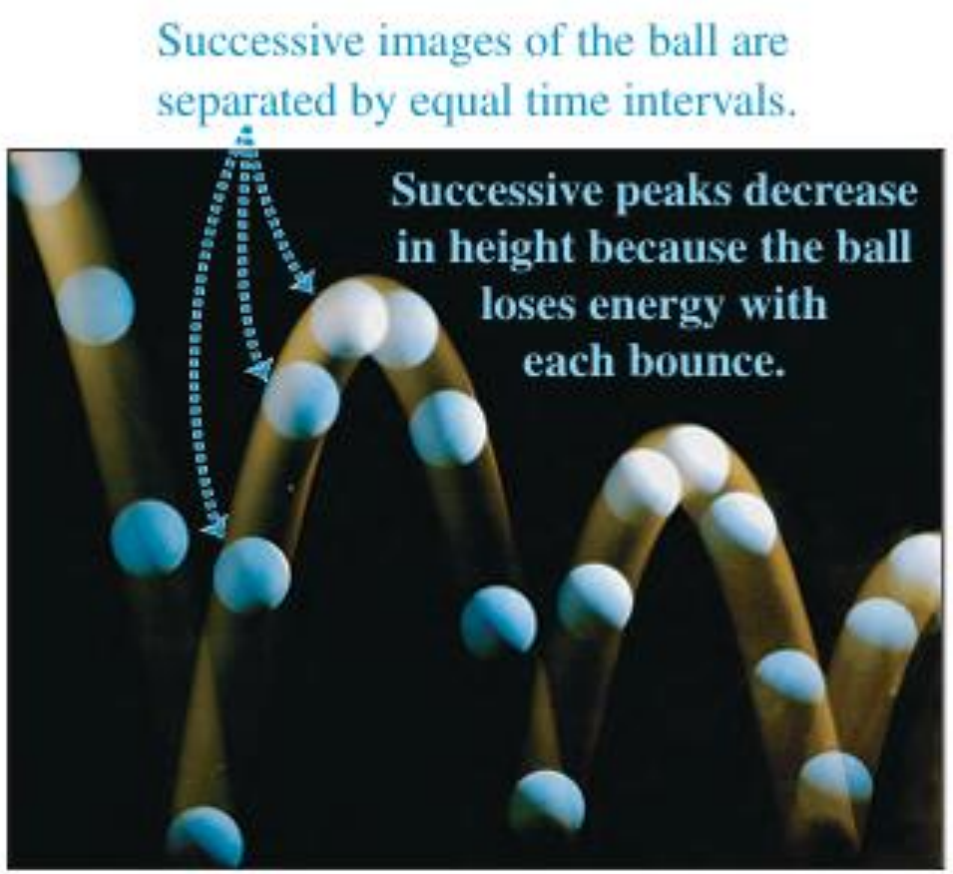
$$R = \frac{v^2{}_o Sin(2\theta)}{g}$$



- Angles that sum to 90° (e.g., 30° and 60°) give the same range
- Higher angle gives longer time of flight and maximum height

Projectile Motion

Figure 3.19 The nearly parabolic trajectories of a bouncing ball.



Example

A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0\ m/s$ at an $angle = 53.1°$.

(a) Find the position of the ball and its velocity (magnitude and direction) at $t = 2.00\ s$.

(b) Find the time when the ball reaches the highest point of its flight, and its height h at this time.

(c) Find the horizontal range R—that is, the horizontal distance from the starting point to where the ball hits the ground—and the ball's velocity just before it hits.

Figure 3.23 Our sketch for this problem.

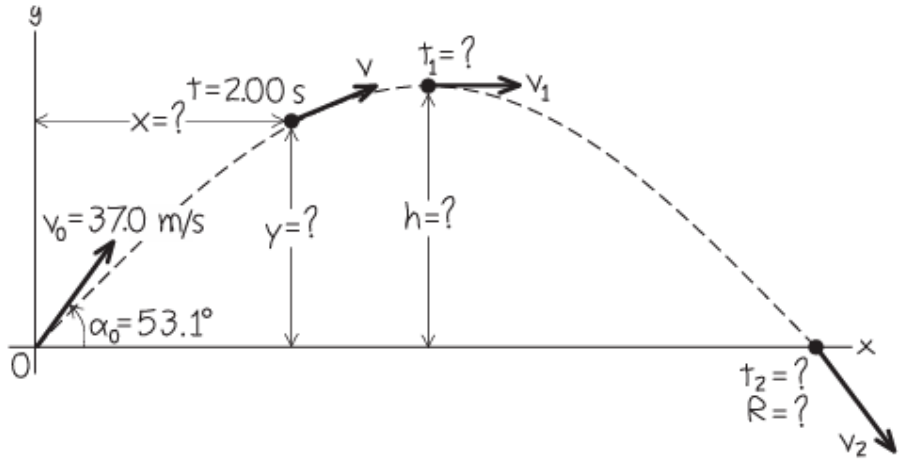
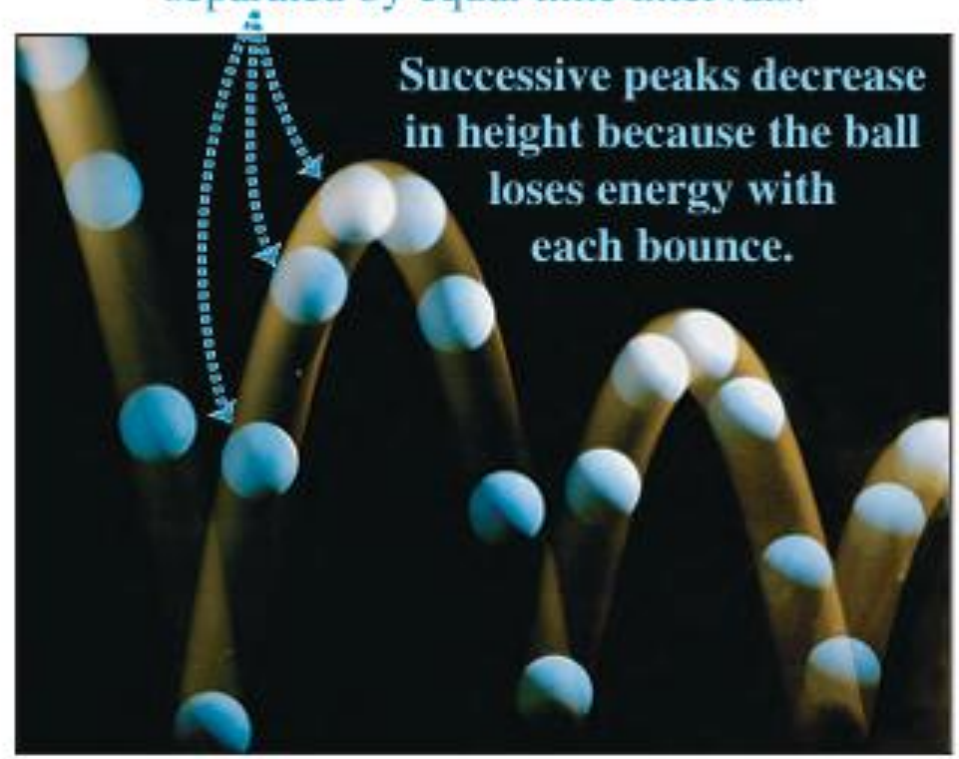


Figure 3.19 The nearly parabolic trajectories of a bouncing ball.



Projectile Motion

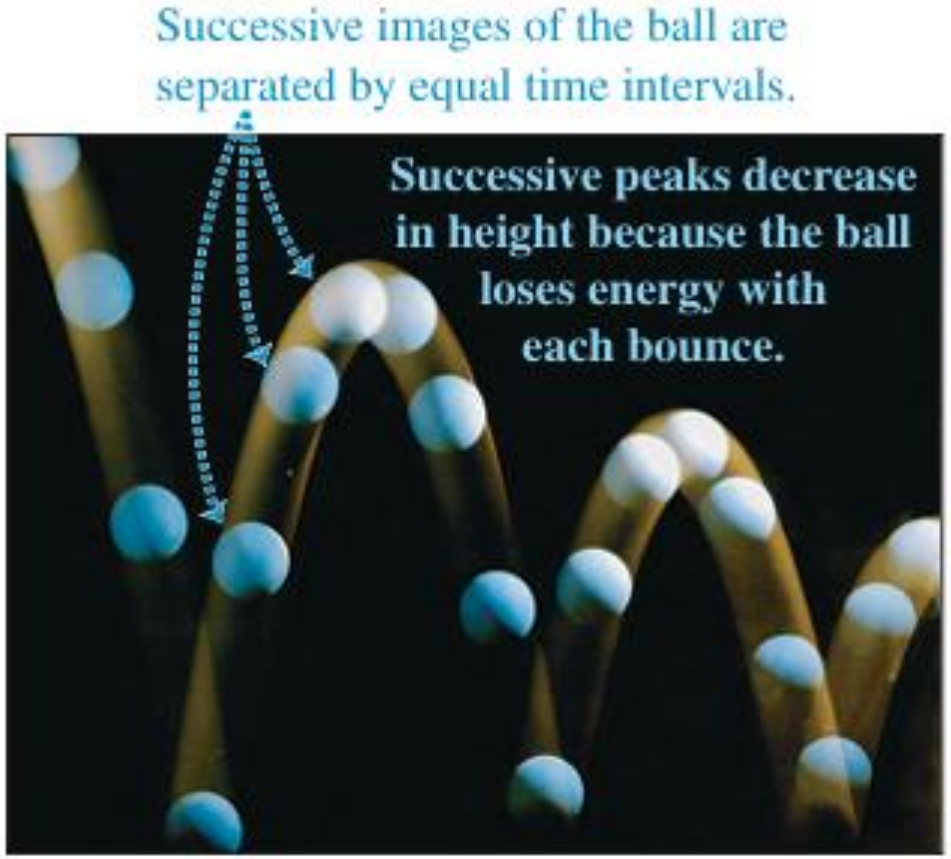
Projectile motion is the motion of an ob
subject only to acceleration due to grav
is called its trajectory, which is always
resistance).

$$\overrightarrow{a_{av}} = \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{\overrightarrow{v_2} - \overrightarrow{v_1}}{t_2 - t_1}$$

$$\overrightarrow{a_{ins}} = \frac{d\boldsymbol{v}}{dt} = \frac{d}{dt}(vx\hat{\imath} + v_y\hat{\jmath} + v_z\hat{k})$$



Figure 3.19 The nearly parabolic trajectories of a bouncing ball.



Example



(c) Find its instantaneous velocity at t_1 = 1.0 s by taking Δt = 0.1 s, then 0.01 s, then 0.001 s.

$\Delta t = 0.1$ s $\quad t_1 = 1\ s \quad t_2 = 1.1\ s$

$x_2 = 20\ m + (5\ m/s^2)(1.1)^2 = 26.05$ m

$V_{av} = \frac{26.05-25}{1.1-1} = 10.5\ m/s$

(d) Derive an expression for the cheetah's instantaneous velocity as a function of time, and use it to find v_x at t = 1.0 s and t = 2.0 s.

$V_x = \frac{dx}{dt} = \frac{d(20\ m+(5\ m/s^2)t^2\ )}{dt} = \left(10.5\frac{m}{s^2}\right)t$

Projectile Motion

$v_{ox} = v_o Cos(\theta)$

$v_{oy} = v_o Sin(\theta)$

$x = v_{0x} t$

$y = v_{0y} t + (1/2)(-g)t^2$

$x = v_o Cos(\theta) t$

$y = v_o Sin(\theta) t + (1/2)(-g)t^2$

$v_x = v_o Cos(\theta)$

$v_y = v_o Sin(\theta) t + (-g)t$

$x/(v_o Cos(\theta)) = t$

$$y = v_o Sin(\theta)[x/(v_o Cos(\theta))] + \left(\frac{1}{2}\right)(-g)[x/(v_o Cos(\theta))]^2$$

Projectile Motion

$$y = v_o Sin(\theta)[x/(v_o Cos(\theta))] + \left(\frac{1}{2}\right)(-g)[x/(v_o Cos(\theta))]^2$$

$$y = x\ tan(\theta) - \left(\frac{g}{2}\right)\left(\frac{g}{(v_o Cos(\theta))^2}\right)[x]^2$$

$$y = x\ b - c[x]^2$$

Equation of a Parabola

Equations of Projectile Motion

Time of Flight:

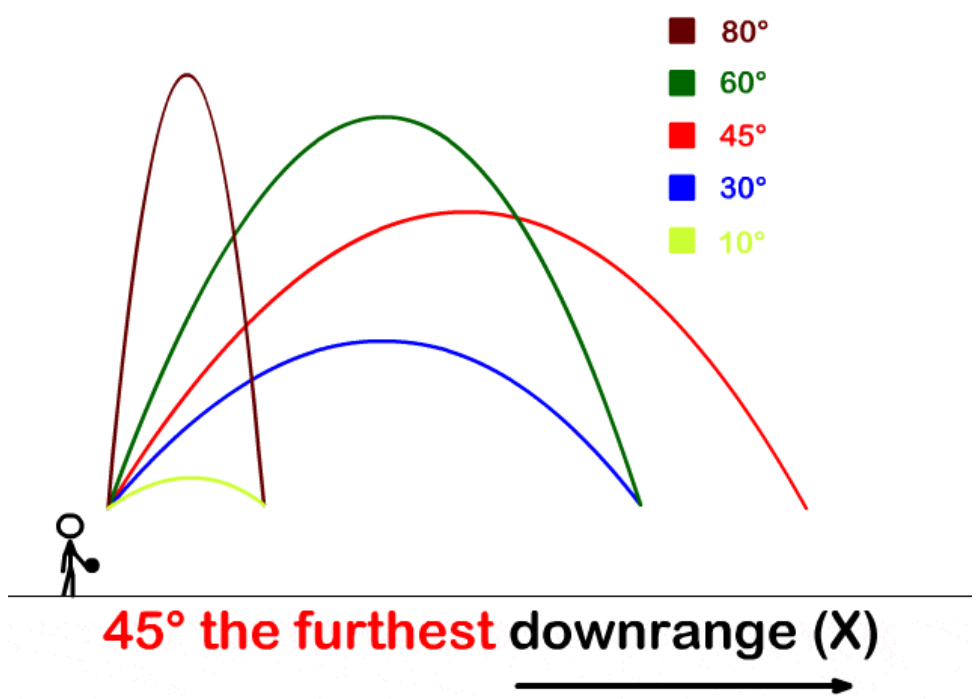
$$T = \frac{2v_o Sin(\theta)}{g}$$

Max. Height:

$$H = \frac{2v^2{}_o Sin^2(\theta)}{2g}$$

Range:

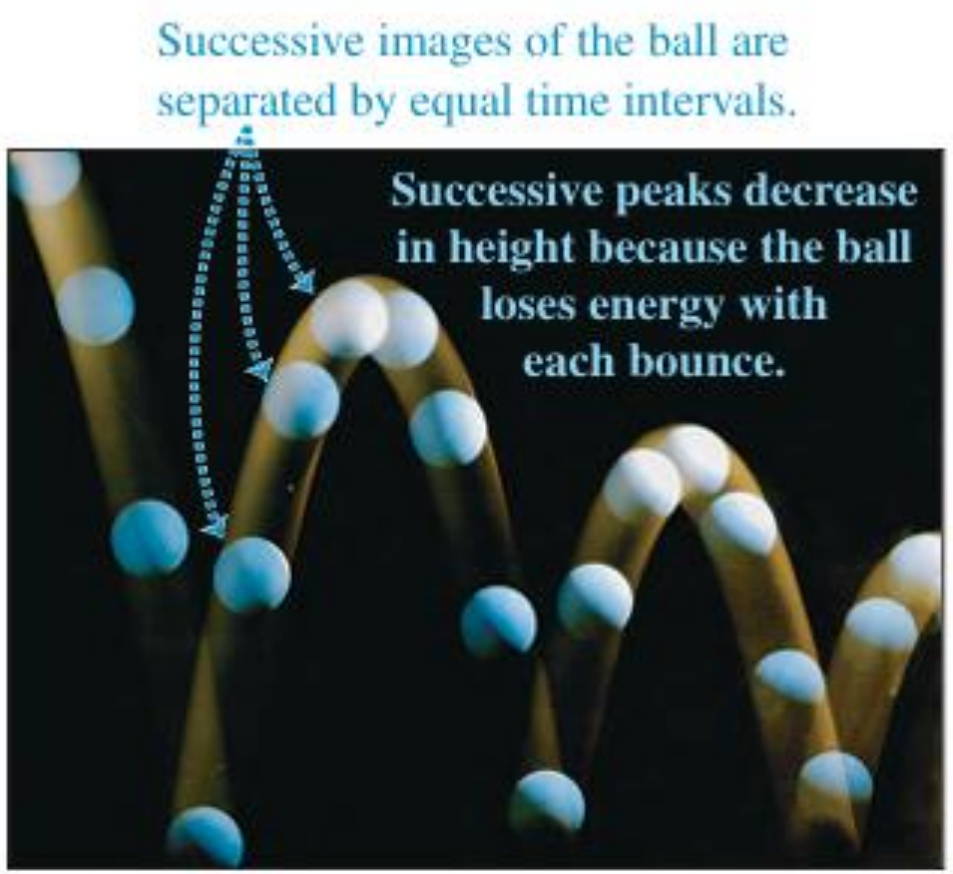
$$R = \frac{v^2{}_o Sin(2\theta)}{g}$$



- Angles that sum to 90° (e.g., 30° and 60°) give the same range
- Higher angle gives longer time of flight and maximum height

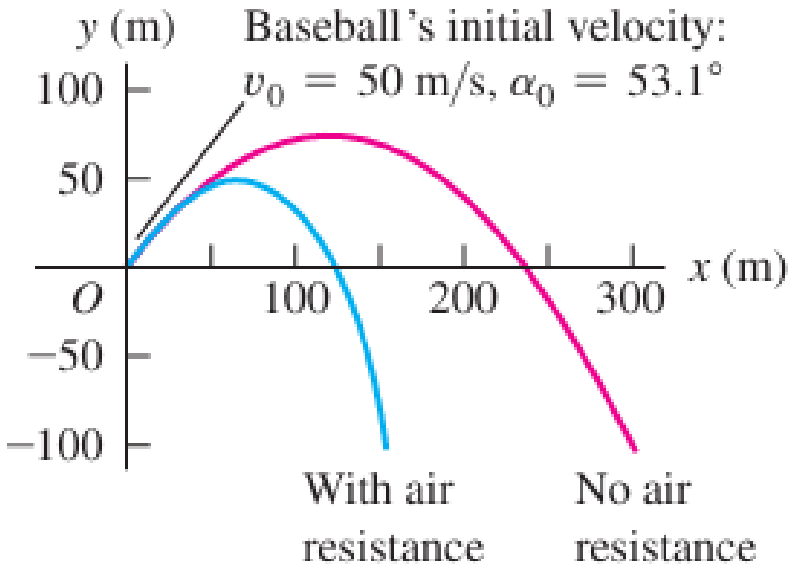
Projectile Motion

Figure 3.19 The nearly parabolic trajectories of a bouncing ball.



Projectile Motion

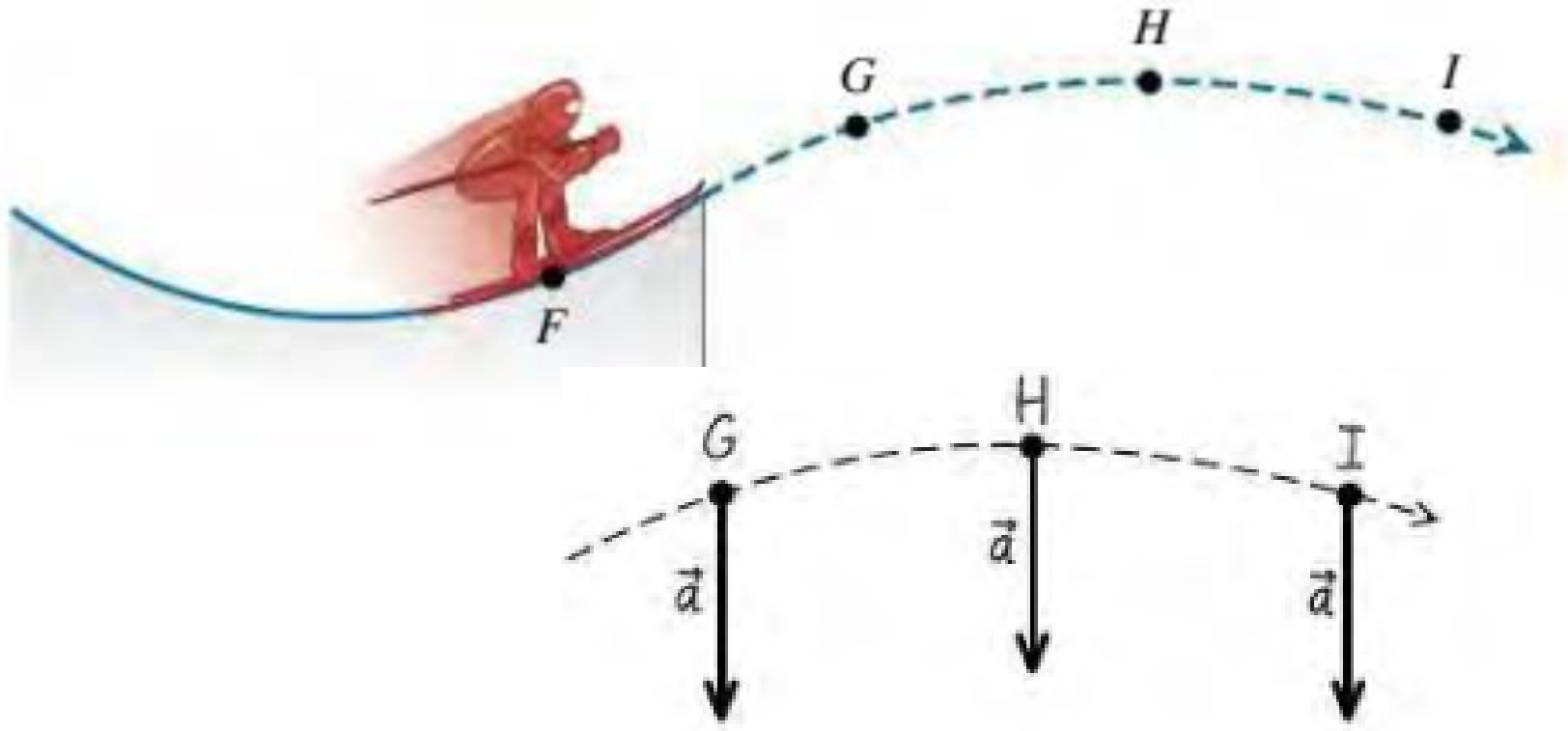
Figure 3.20 Air resistance has a large cumulative effect on the motion of a baseball. In this simulation we allow the baseball to fall below the height from which it was thrown (for example, the baseball could have been thrown from a cliff).





Let's consider the skier.

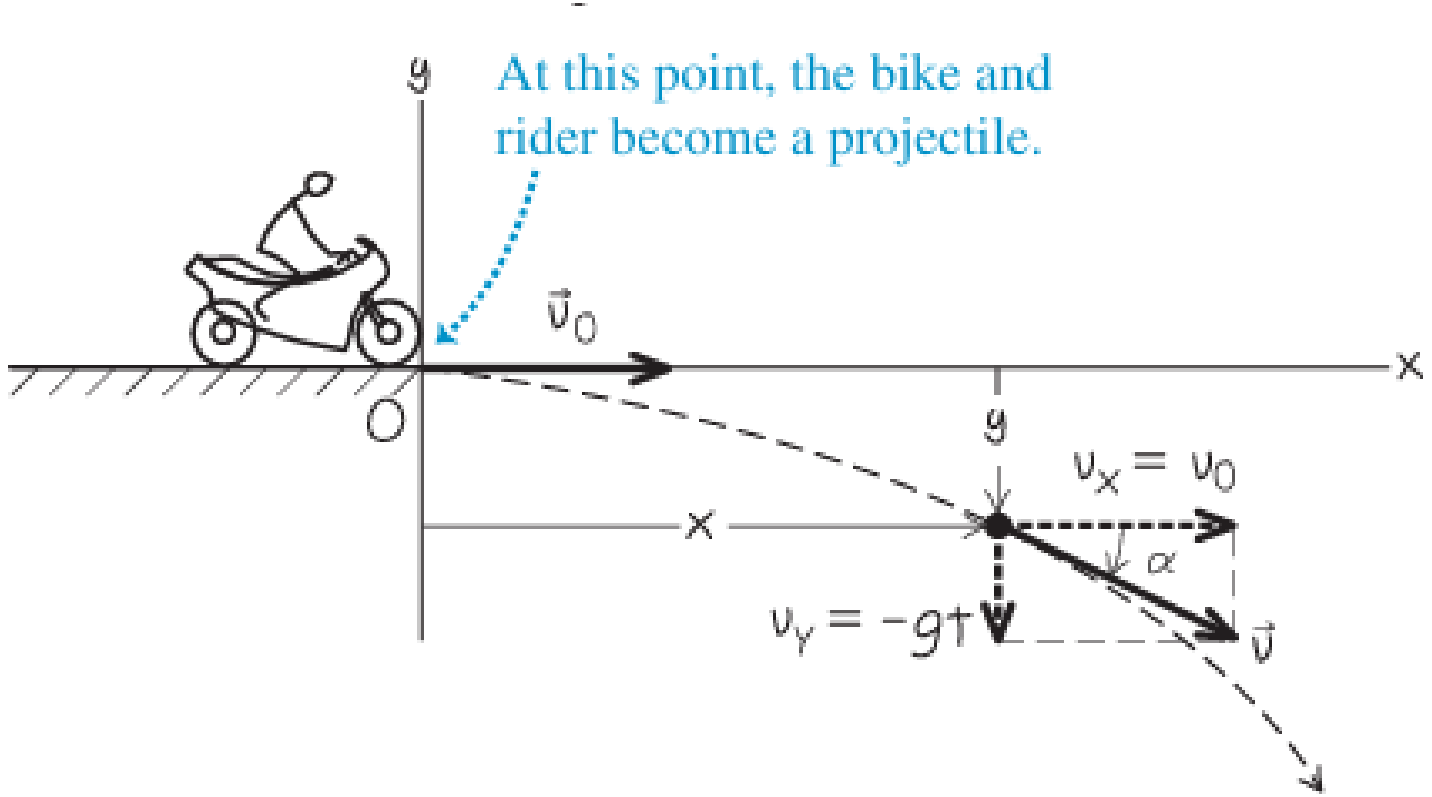
What is his acceleration at each of the points G, H, and I in Fig. after she flies off the ramp? Neglect air resistance.



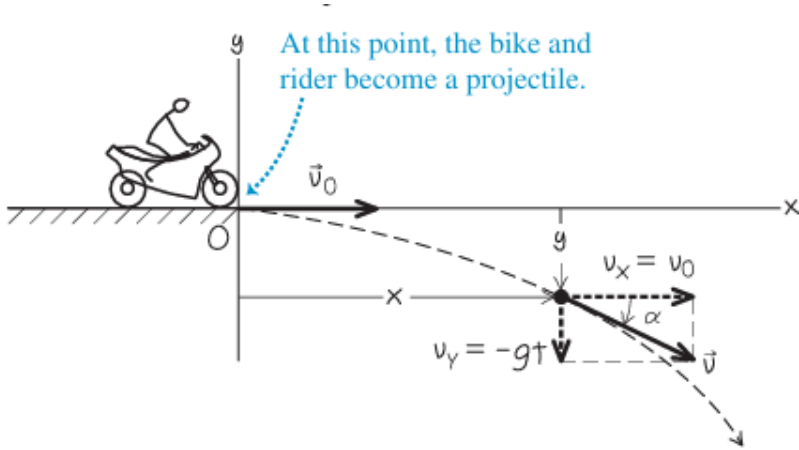
- **Magnitude:** The magnitude of acceleration at G, H, and I is constant and equal to $g \approx 9.80\ \text{m/s}^2$.
- **Direction:** The vector $\vec{a}$ points vertically downward toward the Earth at all three points.

Example

A motorcycle stunt rider rides off the edge of a cliff. Just at the edge his velocity is horizontal, with magnitude 9.0 m/s. Find the motorcycle's **position**, **distance** from the edge of the cliff, and **velocity** 0.50 s after it leaves the edge of the cliff.



Example



A motorcycle stunt rider rides off the edge of a cliff. Just at the ed
with magnitude 9.0 m/s. Find the motorcycle's **position**, **distance** from the edge of the cliff,
and **velocity** 0.50 s after it leaves the edge of the cliff.

$$x(t) = v_{0x}t = 9.0 \times 0.50 = 4.5 \text{ m}$$

$$y(t) = v_{0y}t + \frac{1}{2}a_y t^2$$

$$y(0.50) = 0 + \frac{1}{2}(-9.8)(0.50)^2$$

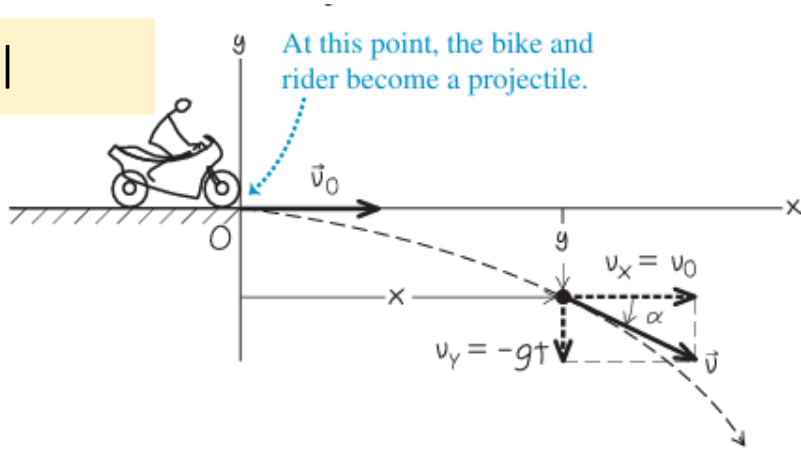
$$= -4.9 \times 0.25 = -1.225 \text{ m}$$

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{(4.5)^2 + (1.225)^2} = \boxed{d \approx 4.66 \text{ m}}$$

velocity is directed $28.6°$ below the horizontal

Example



A motorcycle stunt rider rides off the edge of a cliff. Just at the ed
with magnitude $9.0\ m/s$. Find the motorcycle's **position**, **distance** from the edge of the cliff,
and **velocity** $0.50\ s$ after it leaves the edge of the cliff.

$$x(t) = v_{0x}t = 9.0 \times 0.50 = 4.5\ \text{m}$$

$$v_x = 9.0\ \text{m/s}$$

$$y(t) = v_{0y}t + \frac{1}{2}a_y t^2$$

$$y(0.50) = 0 + \frac{1}{2}(-9.8)(0.50)^2$$

$$= -4.9 \times 0.25 = -1.225\ \text{m}$$

$$v_y(t) = v_{0y} + a_y t = 0 + (-9.8)(0.50) = -4.9\ \text{m/s}$$

$$\boxed{\vec{v} = 9.0\,\hat{\imath} - 4.9\,\hat{\jmath}\ \text{m/s}}$$

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{(4.5)^2 + (1.225)^2} = \boxed{d \approx 4.66\ \text{m}}$$

$$|\,\vec{v}\,| = \sqrt{(9.0)^2 + (-4.9)^2}$$

$$\boxed{|\,\vec{v}\,| \approx 10.2\ \text{m/s}}$$

$$\theta = \tan^{-1}\left(\frac{|\,v_y\,|}{v_x}\right) = \tan^{-1}\left(\frac{4.9}{9.0}\right) = 28.6°$$

Example

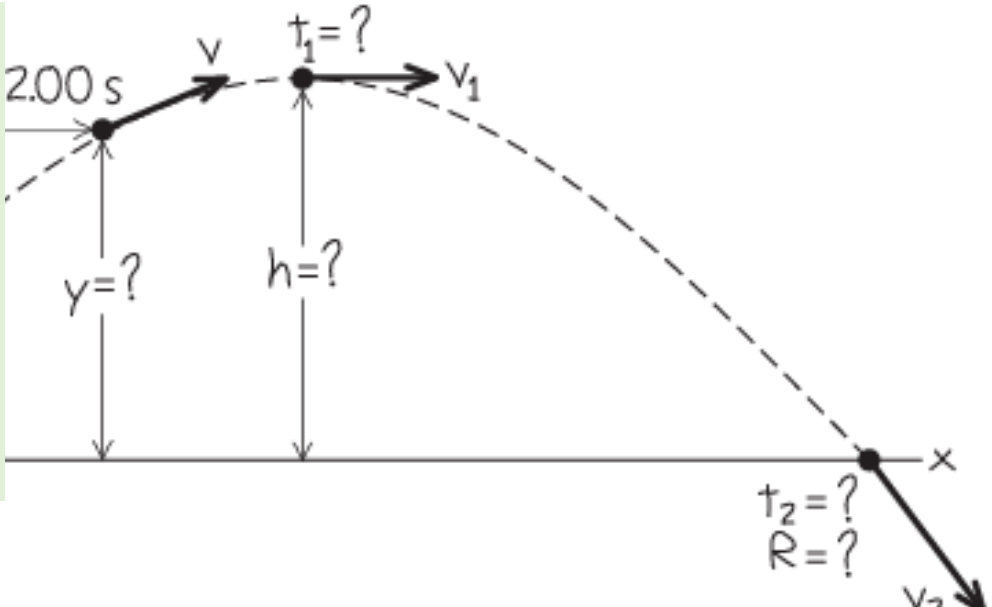
A batter hits a baseball so that it leaves the bat at speed $\boldsymbol{v_0}$ **= 37.0 m/s** at an angle α_0= 53.1°.
(a) Find the position of the ball and its velocity (magnitude and direction) at t = 2.00 s.
(b) Find the time when the ball reaches the highest point of its flight, and its height h at this time. (c) Find the horizontal range R—that is, the horizontal distance from the starting point to where the ball hits the ground—and the ball's velocity just before it hits.

Initial velocity components:

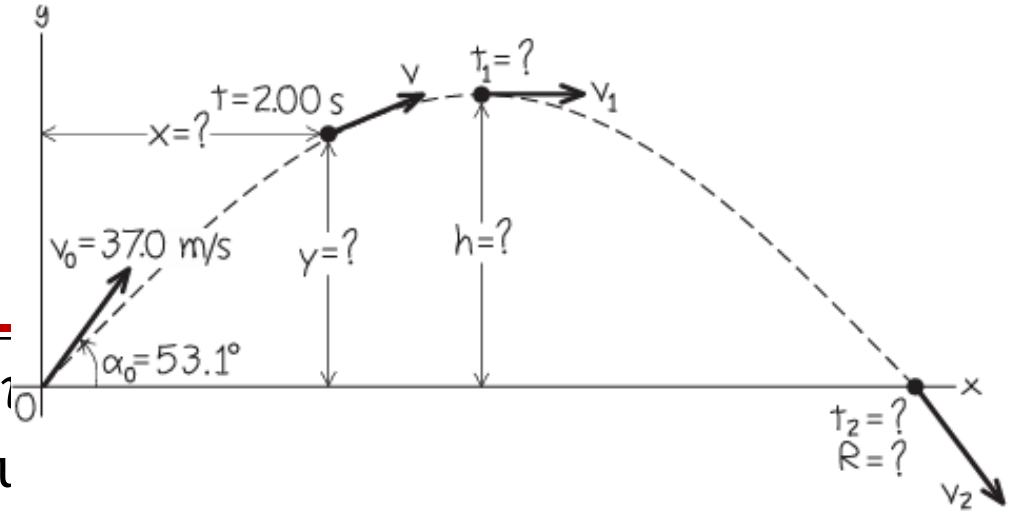
$$v_{0x} = v_0 \cos\alpha_0 = 37.0 \cos 53.1° = 22.2 \text{ m/s}$$
$$v_{0y} = v_0 \sin\alpha_0 = 37.0 \sin 53.1° = 29.6 \text{ m/s}$$

Acceleration:

$$a_x = 0, a_y = -g = -9.8 \text{ m/s}^2$$



Example



A batter hits a baseball so that it leaves the bat at speed 1
(a) Find the position of the ball and its velocity (magnitu
(b) Find the time when the ball reaches the highest point of its flight, and its height h at this time. (c) Find the horizontal range R—that is, the horizontal distance from the starting point to where the ball hits the ground—and the ball's velocity just before it hits.

$$x(t) = v_{0x}t = 22.2 \times 2.00 = 44.4\ \text{m}$$

$$y(t) = v_{0y}t + \frac{1}{2}a_y t^2$$

$$y(2) = 29.6(2.00) - \frac{1}{2}(9.8)(2.00)^2$$

$$= 59.2 - 4.9(4) = 59.2 - 19.6 = 39.6\ \text{m}$$

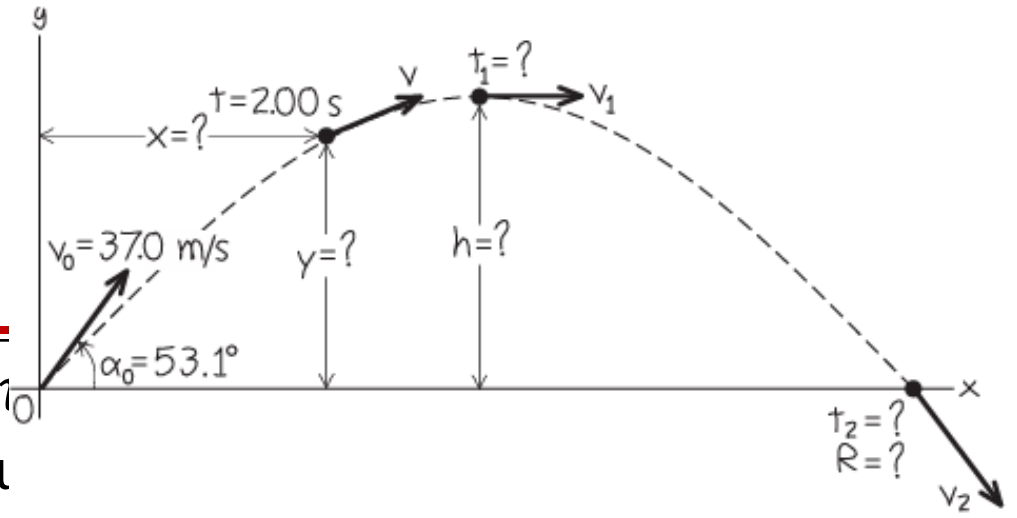
$$\boxed{(44.4,\ 39.6)\ \text{m}}$$

$$v_x = 22.2\ \text{m/s}$$

$$v_y(t) = v_{0y} + a_y t = 29.6 - 9.8(2.00)$$
$$= 29.6 - 19.6 = 10.0\ \text{m/s}$$

$$\vec{v} = 22.2\hat{\imath} + 10.0\hat{\jmath}\ \text{m/s}$$

Example



A batter hits a baseball so that it leaves the bat at speed 1
(a) Find the position of the ball and its velocity (magnitu
(b) Find the time when the ball reaches the highest point of its flight, and its height h at this time. (c) Find the horizontal range R—that is, the horizontal distance from the starting point to where the ball hits the ground—and the ball's velocity just before it hits.

$$v_y(t) = v_{0y} - gt = 0$$

$$t = \frac{v_{0y}}{g} = \frac{29.6}{9.8} \approx 3.02 \text{ s}$$

$$h = y(t) = v_{0y}t - \frac{1}{2}gt^2$$

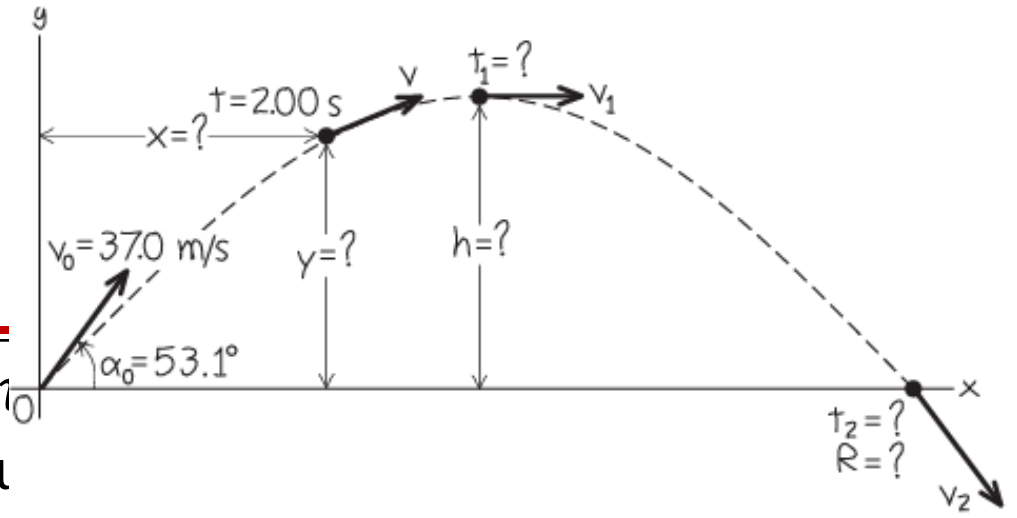
$$h = 29.6(3.02) - 4.9(3.02)^2$$

$$= 89.39 - 4.9(9.1204) = 89.39 - 44.69$$

$$\approx 44.7 \text{ m}$$

$$\boxed{t \approx 3.02 \text{ s},\ h \approx 44.7 \text{ m}}$$

Example



A batter hits a baseball so that it leaves the bat at speed
(a) Find the position of the ball and its velocity (magnitu
(b) Find the time when the ball reaches the highest point of its flight, and its height h at this time. (c) Find the horizontal range R—that is, the horizontal distance from the starting point to where the ball hits the ground—and the ball's velocity just before it hits.

$$T = \frac{2v_{0y}}{g} = \frac{2(29.6)}{9.8} = \frac{59.2}{9.8} \approx 6.04 \text{ s}$$

$$R = v_{0x}T = 22.2 \times 6.04 \approx 134.1 \text{ m}$$

$$R = \frac{v_0^2 \sin 2\alpha_0}{g}$$

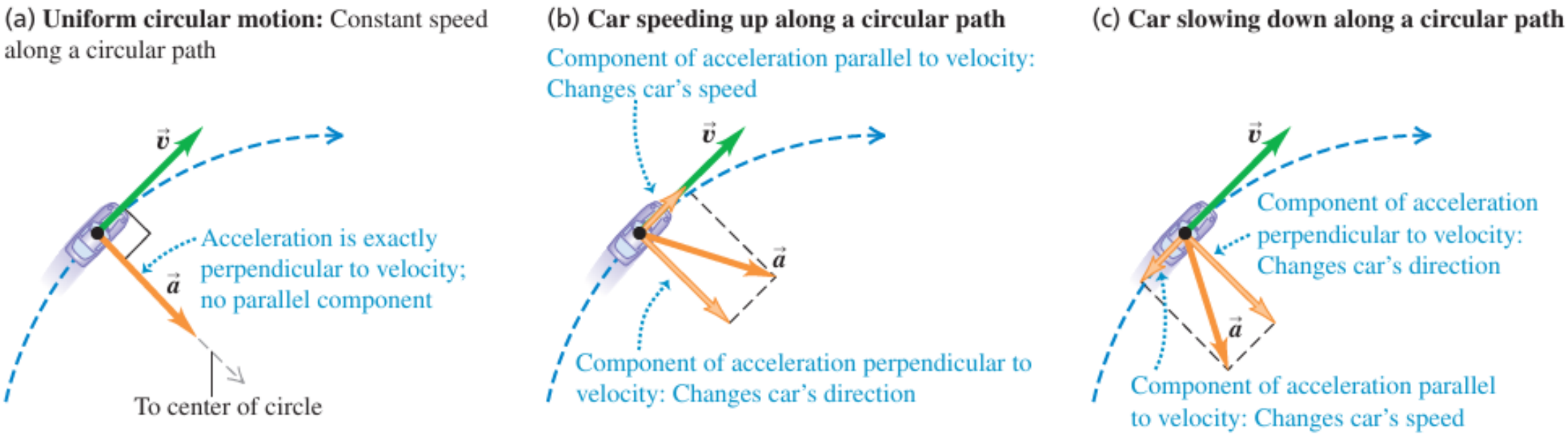
$$\sin(2 \times 53.1°) = \sin 106.2° \approx 0.960$$

$$R = \frac{(37.0)^2(0.960)}{9.8} = \frac{1369 \times 0.960}{9.8}$$

$$\approx 134.1 \text{ m}$$

Motion in a Circle

When a particle moves along a curved path, the direction of its velocity changes.



the particle must have a component of acceleration perpendicular to the path,

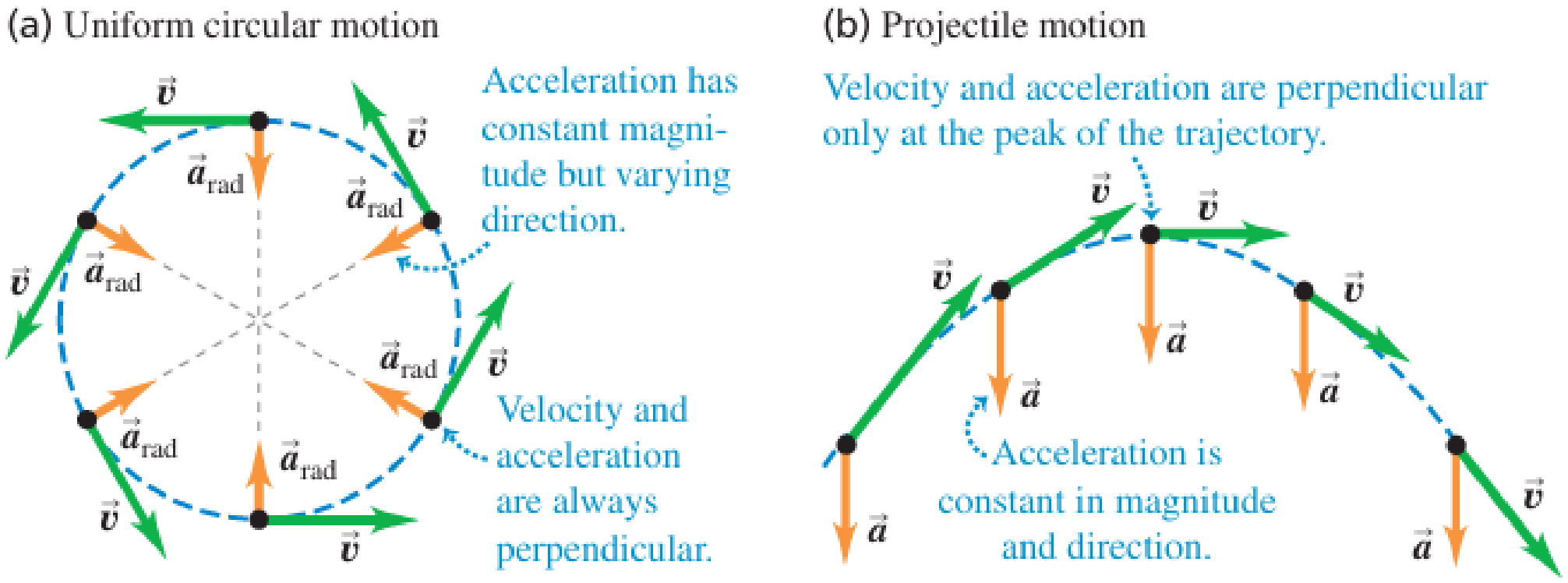
Uniform Circular Motion

When a particle moves in a circle with constant speed, the motion is called uniform circular motion.

- A car rounding a curve with constant radius at constant speed,
- A satellite moving in a circular orbit, and
- An ice skater skating in a circle with constant speed

In uniform circular motion, an object moves along a circular path with constant speed. Despite the constant speed, the object experiences acceleration because its velocity vector continuously changes direction. This acceleration is called centripetal acceleration and is always directed toward the center of the circle.

Uniform Circular Motion



Uniform Circular Motion

$$\frac{|\Delta\bar{v}|}{v_1} = \frac{\Delta s}{R}$$

$$|\Delta\bar{v}| = v_1 \frac{\Delta s}{R}$$

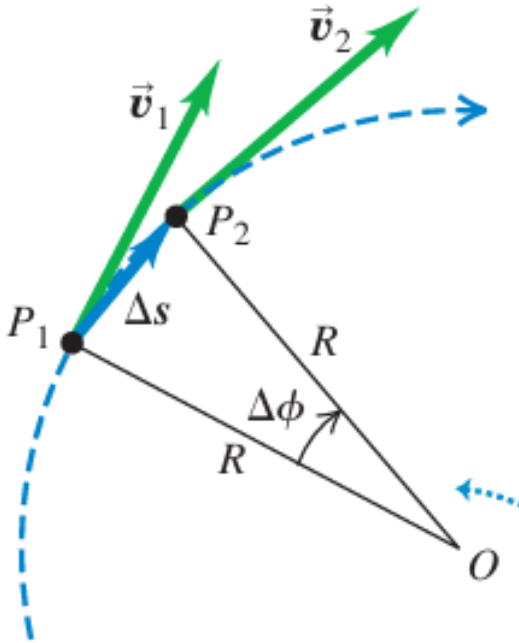
$$\frac{|\Delta\bar{v}|}{\Delta t} = \frac{v_1}{R}\frac{\Delta s}{\Delta t}$$

$$\lim_{t\to 0}\frac{|\Delta\bar{v}|}{\Delta t} = \frac{v_1}{R}\lim_{t\to 0}\frac{\Delta s}{\Delta t}$$

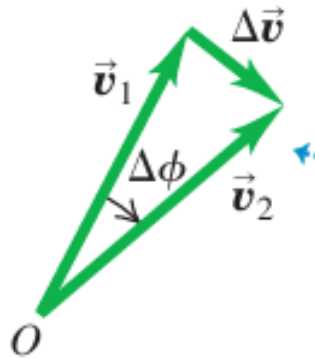
$$a = \frac{v^2}{R}$$

$$a_{rad} = \frac{v^2}{R}$$

(a) A particle moves a distance Δs at constant speed along a circular path.



(b) The corresponding change in velocity $\Delta\vec{v}$. The average acceleration is in the same direction as $\Delta\vec{v}$



These two triangles are similar.

Example

An Aston Martin V12 Vantage sports car has a "lateral acceleration" of 0.97g = (0.97)(9.8 m/s^2) = 9.5 m/s^2.
This is the maximum centripetal acceleration the car can sustain without skidding out of a curved path.
If the car is traveling at a constant $40\ m/s$ (about 89 mi/h, or 144 km/h) on level ground, what is the radius R of the tightest un-banked curve it can negotiate?

Circular motion: $a_c = \frac{v^2}{R}$

We need the **tightest curve** (smallest radius R) at maximum centripetal acceleration:

$$R = \frac{v^2}{a_c}$$

$$R = \frac{v^2}{a_c} = \frac{(40)^2}{9.5} = 168.42$$

Non-uniform Circular Motic

$$a_{rad} = \frac{v^2}{R}$$

$$\overline{a_{tan}} = \frac{|d\bar{v}|}{dt}$$

APPLICATION Watch Out: Tight Curves Ahead! These roller coaster cars are in nonuniform circular motion: They slow down and speed up as they move around a vertical loop. The large accelerations involved in traveling at high speed around a tight loop mean extra stress on the passengers' circulatory systems, which is why people with cardiac conditions are cautioned against going on such rides.

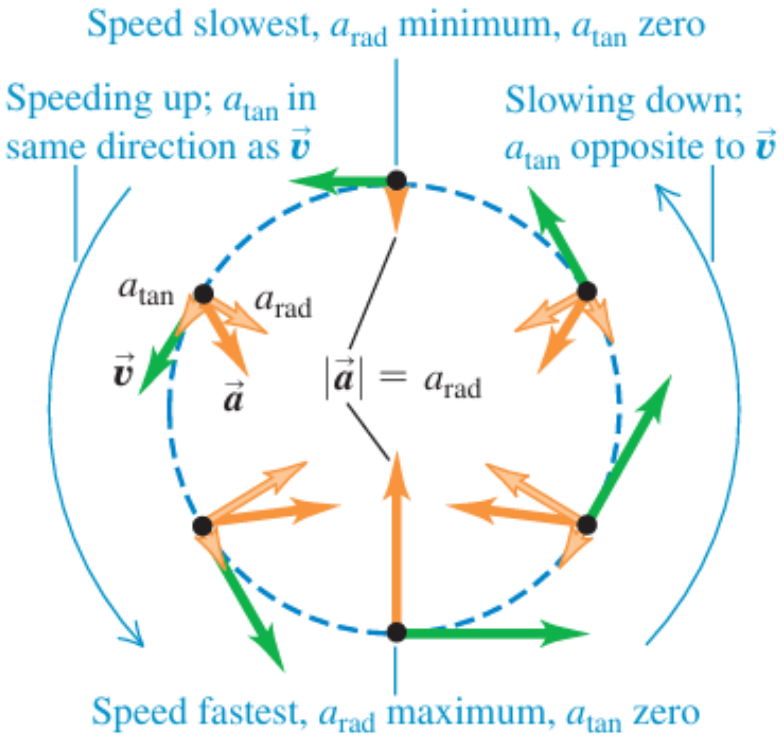


Non-uniform Circular Motion

$$a_{rad} = \frac{v^2}{R}$$

$$\overline{a_{tan}} = \frac{|d\bar{v}|}{dt}$$

Figure 3.30 A particle moving in a vertical loop with a varying speed, like a roller coaster car.



Relative Velocity

when two observers measure the velocity of the same object, they get different results if one observer is moving relative to the other. The velocity seen by a particular observer is called the velocity relative to that observer, or simply relative velocity.

Figure 3.31 Airshow pilots face a complicated problem involving relative velocities. They must keep track of their motion relative to the air (to maintain enough airflow over the wings to sustain lift), relative to each other (to keep a tight formation without colliding), and relative to their audience (to remain in sight of the spectators).



Relative Velocity

the symbol A for the cyclist's frame of reference (at rest with respect to the ground) and the symbol B for the frame of reference of the moving train.

Figure 3.31 Airshow pilots face a complicated problem involving relative velocities. They must keep track of their motion relative to the air (to maintain enough airflow over the wings to sustain lift), relative to each other (to keep a tight formation without colliding), and relative to their audience (to remain in sight of the spectators).



Relative Velocity

the symbol A for the cyclist's frame of reference (at rest with respect to the ground) and the symbol B for the frame of reference of the moving train.

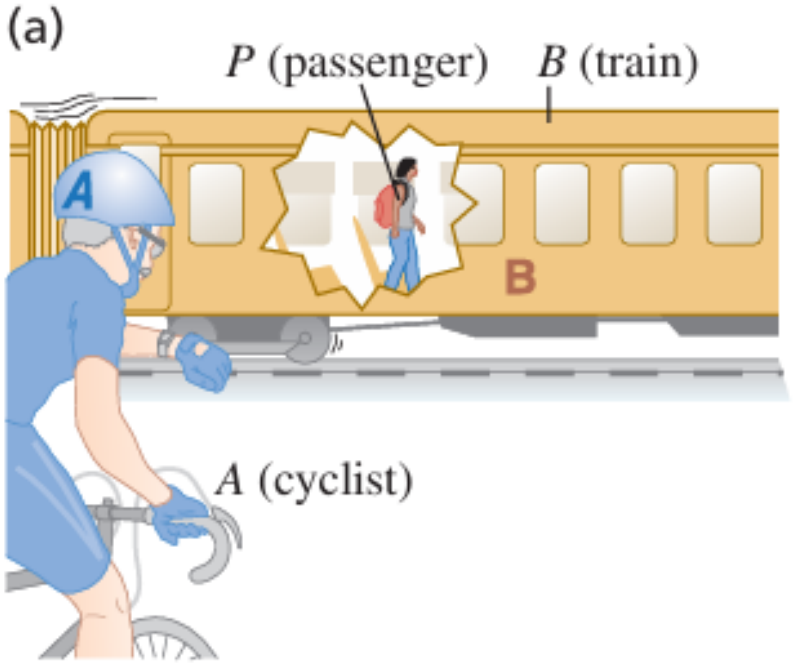
In straight-line motion the position of a point P relative to frame A is given by $x_{P/A}$

$$x_{P/A} = x_{P/B} + x_{B/A}$$

Relative velocity along a line: $v_{P/A\text{-}x} = v_{P/B\text{-}x} + v_{B/A\text{-}x}$

- $v_{P/A\text{-}x}$: *x*-velocity of *P* relative to *A*
- $v_{P/B\text{-}x}$: *x*-velocity of *P* relative to *B*
- $v_{B/A\text{-}x}$: *x*-velocity of *B* relative to *A*

Figure 3.32 (a) A passenger walking in a train. (b) The position of the passenger relative to the cyclist's frame of reference and the train's frame of reference.



Relative Velocity

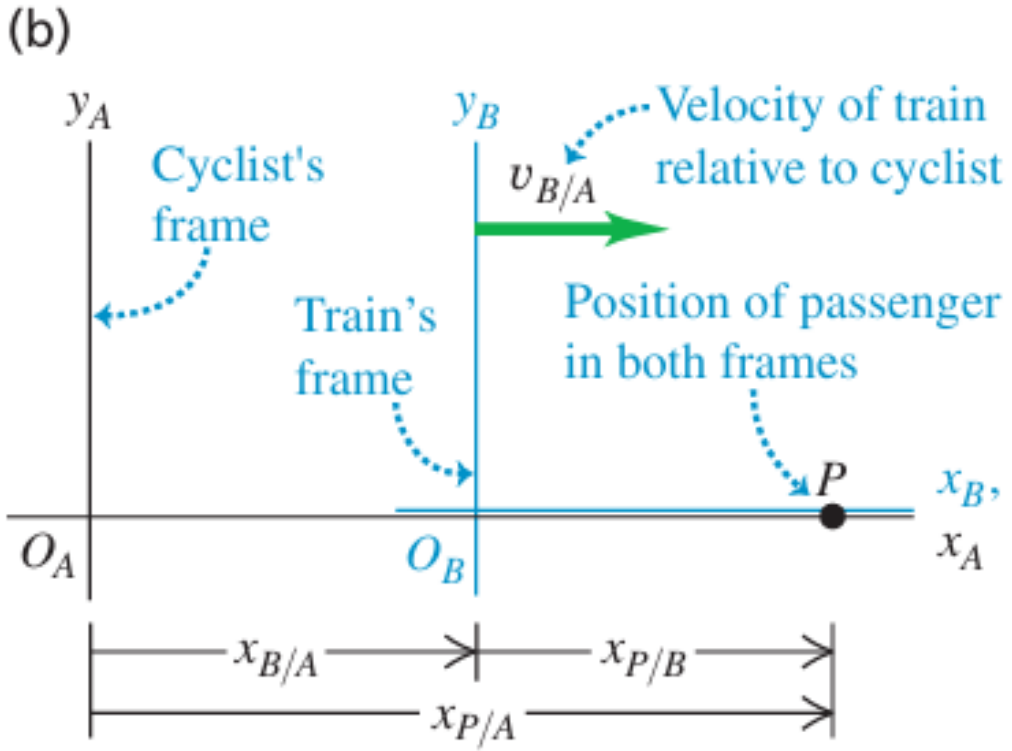
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- $v_{P/A\text{-}x}$: x-velocity of P relative to A
- $v_{P/B\text{-}x}$: x-velocity of P relative to B
- $v_{B/A\text{-}x}$: x-velocity of B relative to A



Relative Velocity

$$x_{P/A} = x_{P/B} + x_{B/A}$$

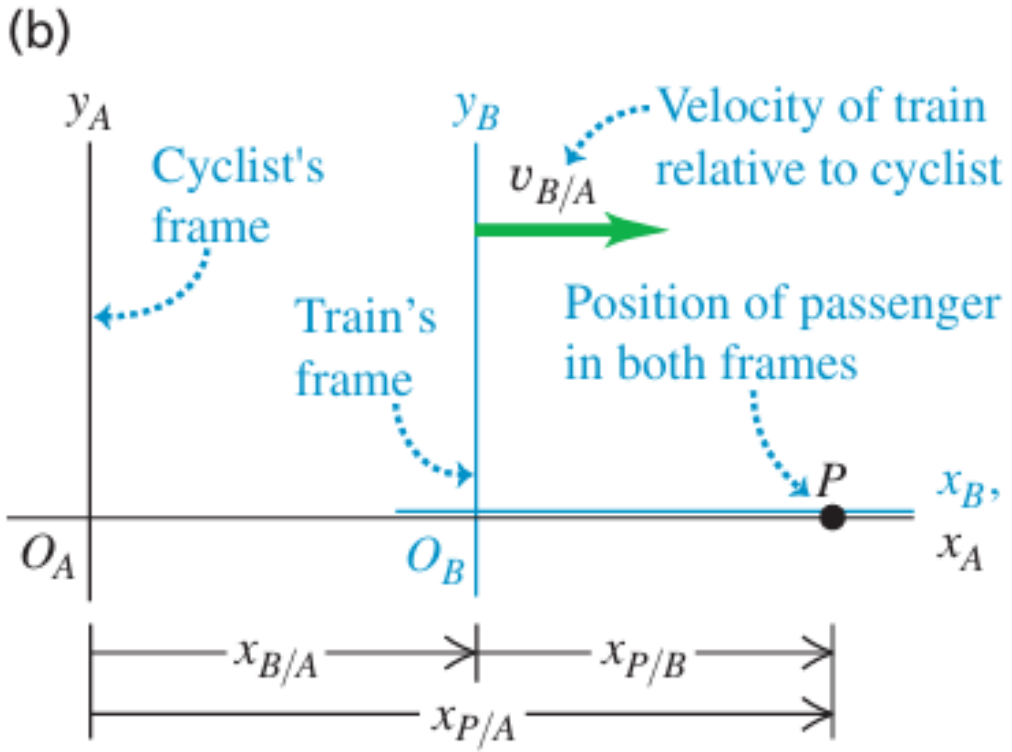
Relative velocity along a line: $v_{P/A\text{-}x} = v_{P/B\text{-}x} + v_{B/A\text{-}x}$

- $v_{P/A\text{-}x}$: *x*-velocity of *P* relative to *A*
- $v_{P/B\text{-}x}$: *x*-velocity of *P* relative to *B*
- $v_{B/A\text{-}x}$: *x*-velocity of *B* relative to *A*

$$v_{P/B\text{-}x} = +1.0 \text{ m/s} \qquad v_{B/A\text{-}x} = +3.0 \text{ m/s}$$

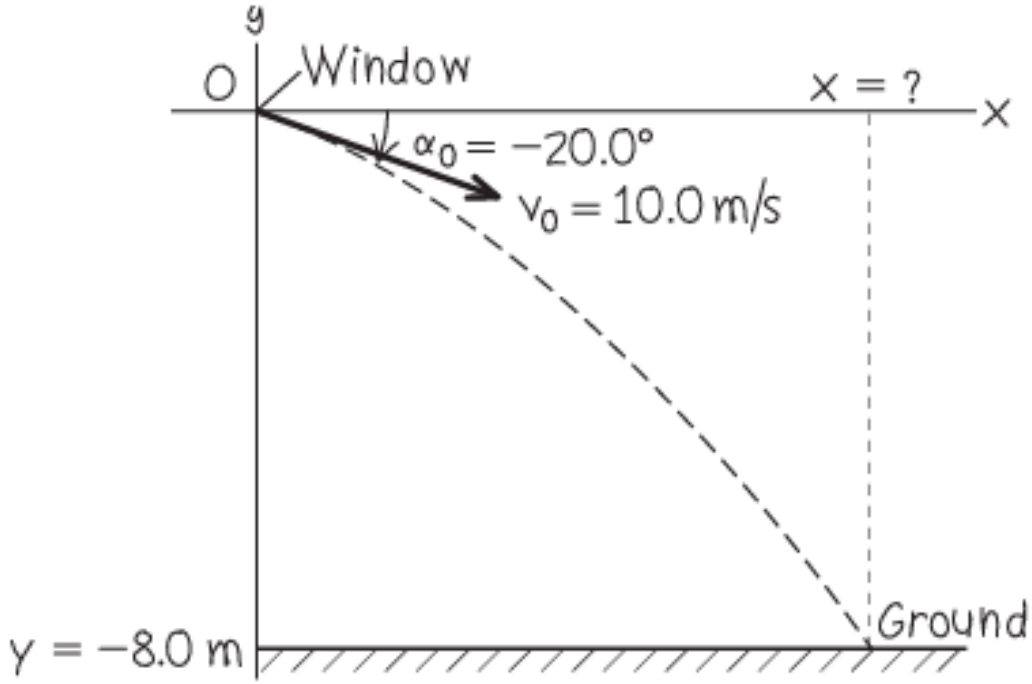
$$v_{P/A\text{-}x} = +1.0 \text{ m/s} + 3.0 \text{ m/s} = +4.0 \text{ m/s}$$

$$v_{P/A\text{-}x} = -1.0 \text{ m/s} + 3.0 \text{ m/s} = +2.0 \text{ m/s}$$

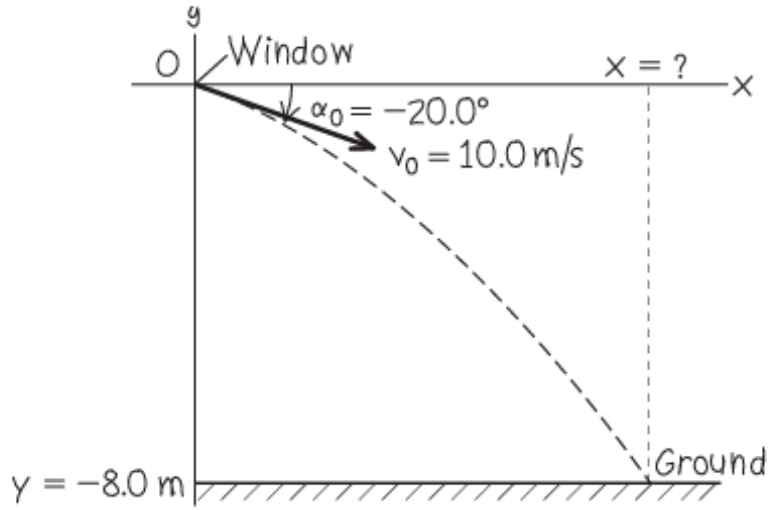


Example

You throw a ball from your window 8.0 m above the ground. When the ball leaves your hand, it is moving at 10.0 m/s at an angle of 20° below the horizontal. How far horizontally from your window will the ball hit the ground? Ignore air resistance.



Example



You throw a ball from your window 8.0 m above the gr
your hand, it is moving at 10.0 m/s at an angle of 20° b
far horizontally from your window will the ball hit the ground? Ignore air resistance.

Height of window	h = 8.0 m
Initial speed	v_0 = 10.0 m/s
Angle below horizontal	θ = 20°
Acceleration due to gravity	g = 9.8 m/s²
Air resistance	Ignored

Horizontal distance(range) from the base of the window to the impact point=?

Initial velocity components:

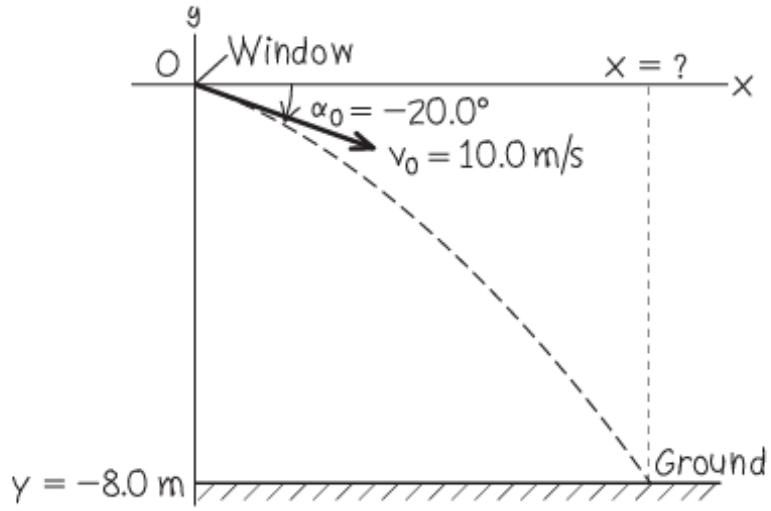
$$v_{0x} = v_0 \cos \alpha_0 = 10 \cos 20° = 9.39 \text{ m/s}$$

$$v_{0y} = v_0 \sin \alpha_0 = 10 \sin 20° = 3.4 \text{ m/s}$$

Acceleration:

$$a_x = 0, a_y = -g = -9.8 \text{ m/s}^2$$

Example



You throw a ball from your window 8.0 m above the gr
your hand, it is moving at 10.0 m/s at an angle of 20° b
far horizontally from your window will the ball hit the ground? Ignore air resistance.

"Horizontal distance"" (range) from the base of the window to the impact point."

$$\mathrm{R} = x(t) = v_{0x}t = 9.39 \times \mathrm{t} = 9.39\mathrm{t}\ \mathrm{m}$$

$$\mathrm{R} = v_{0x}t = 9.39 \times 0.976 = 9.17\ \mathrm{m}$$

$$y(t) = v_{0y}t + \frac{1}{2}a_y t^2$$

$$h = v_{0y}t + \frac{1}{2}gt^2$$

$$h = 8 = 3.4\,t + \frac{1}{2}(9.8)t^2$$

$$t = 0.976\ \mathrm{s}$$

Uniform Circular Motion

When a particle moves in a circle with constant speed, the motion is called uniform circular motion.

A car rounding a curve with constant radius at constant speed, a satellite moving in a circular orbit, and an ice skater skating in a circle with constant speed

In uniform circular motion, an object moves along a circular path with constant speed. Despite the constant speed, the object experiences acceleration because its velocity vector continuously changes direction. This acceleration is called centripetal acceleration and is always directed toward the center of the circle.

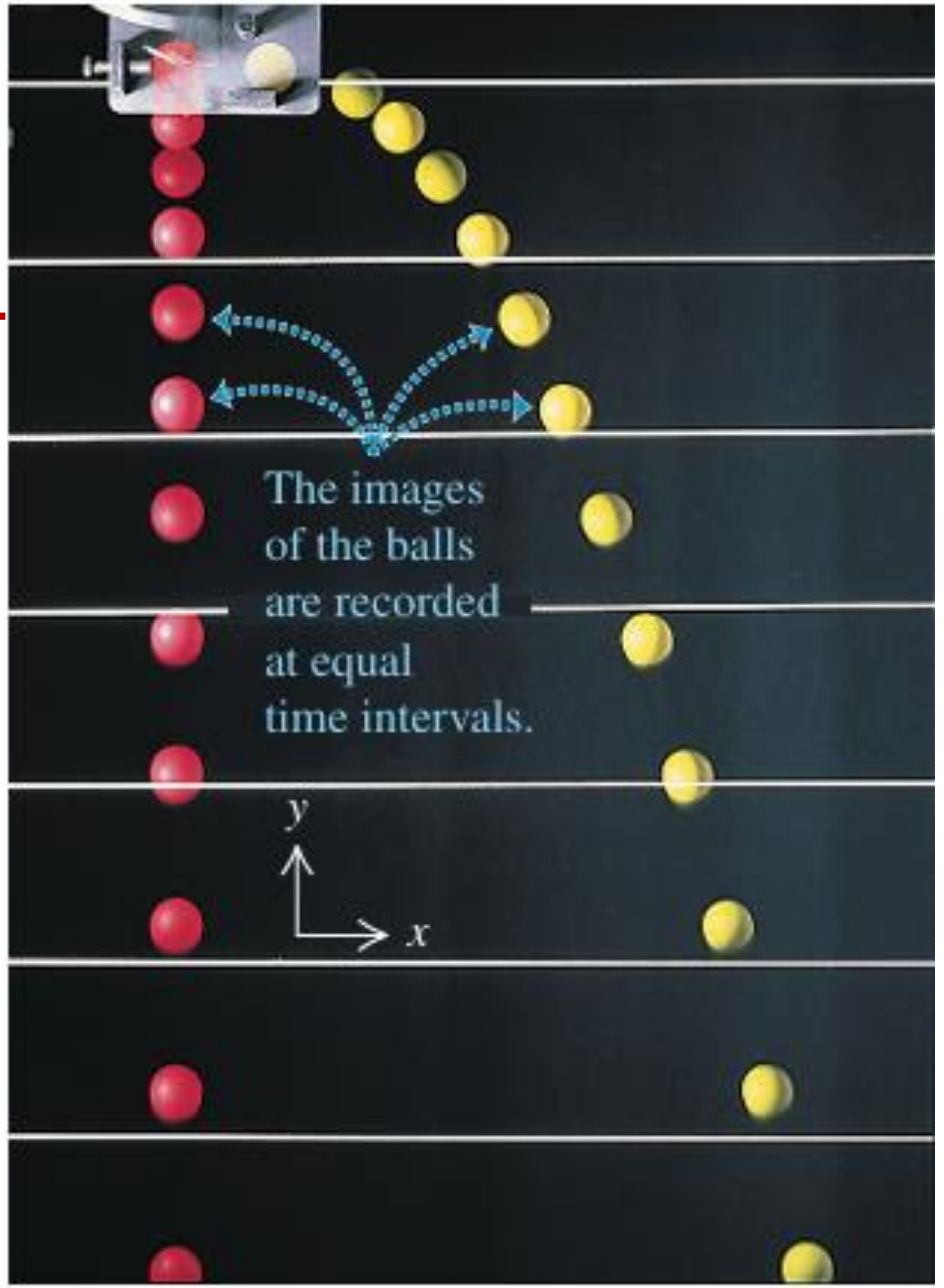
Projectile Motion

Two projectiles: a red ball dropped from rest and a yellow ball projected horizontally from the same height.

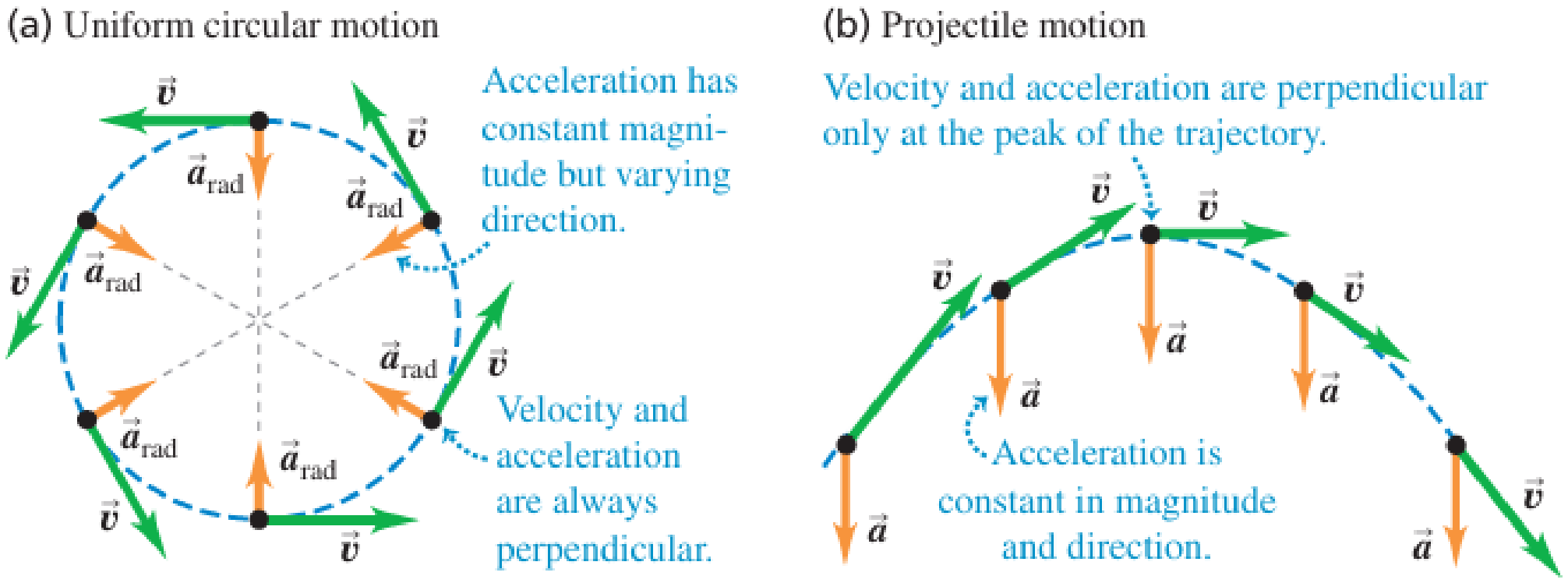
Horizontal motion of the yellow projectile has no effect on its vertical motion.

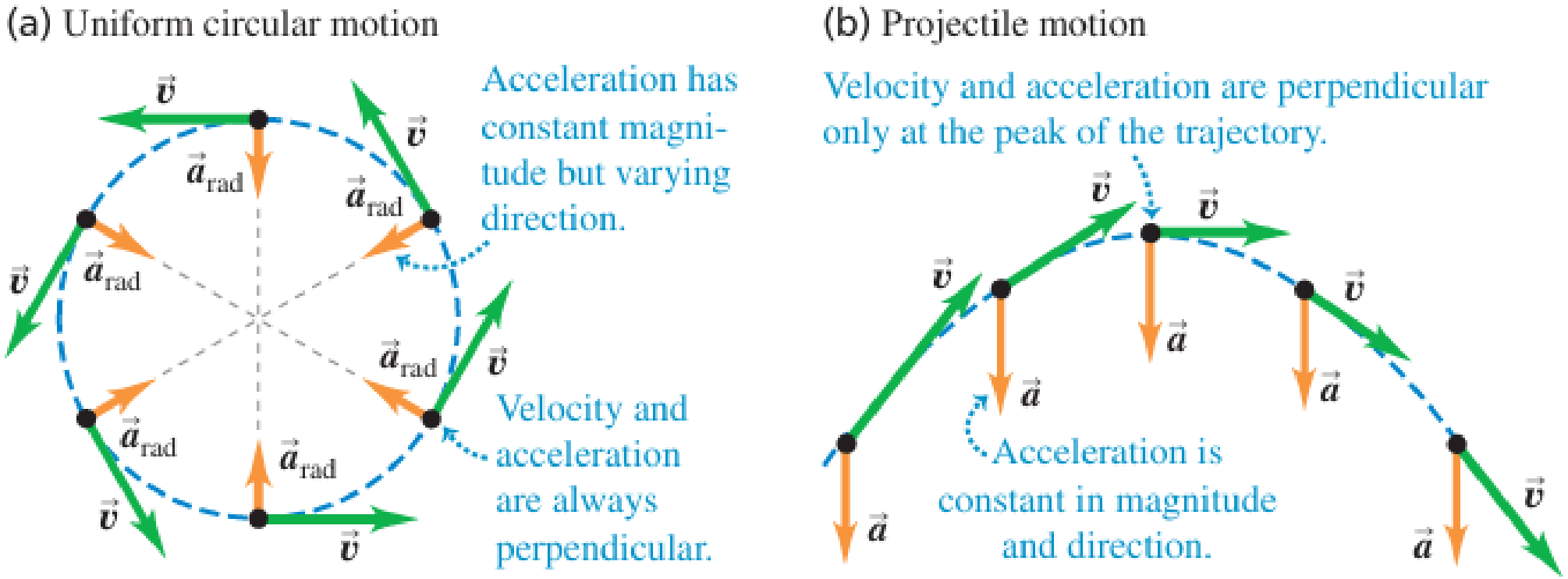
For both projectiles, the x-component of acceleration is zero and the y-component is constant and equal to -g

Projectile motion as a combination of horizontal motion with constant velocity and vertical motion with constant acceleration



Uniform Circular Motion





$$\overrightarrow{a_{av}} = \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{\overrightarrow{v_2} - \overrightarrow{v_1}}{t_2 - t_1}$$

$$\overrightarrow{a_{ins}} = \lim_{n \to 0} \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{d\boldsymbol{v}}{dt}$$

$$\overrightarrow{a_{ins}} = \frac{d\boldsymbol{v}}{dt} = \frac{d}{dt}(v_x\hat{\imath} + v_y\hat{\jmath} + v_z\hat{k}) = a_x\hat{\imath} + a_y\hat{\jmath} + a_z\hat{k}$$

Monkey and Hunter Problem

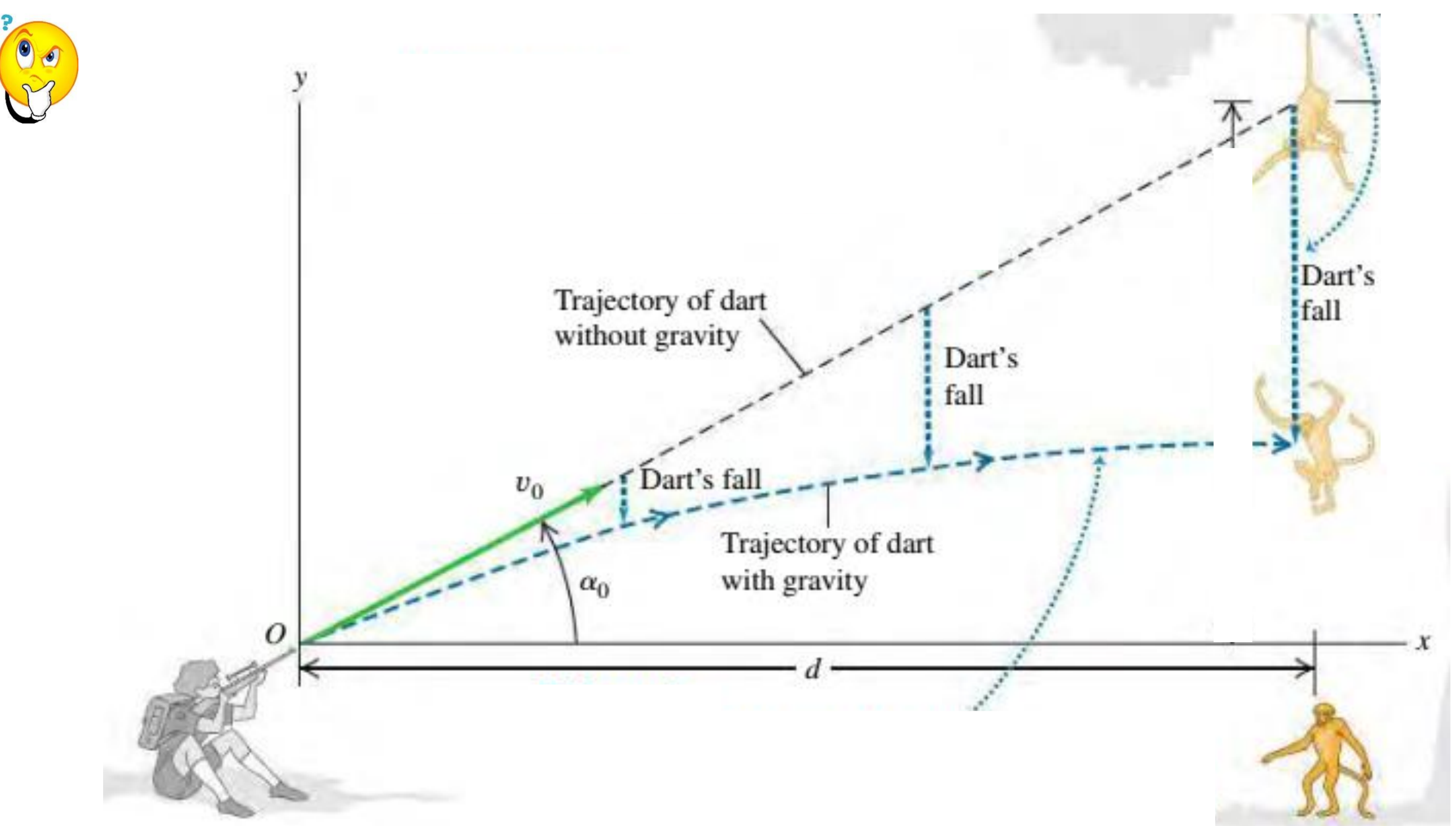
A monkey escapes from the zoo and climbs a tree. After failing to entice the monkey down, the zookeeper fires a tranquilizer dart directly at the monkey. The monkey lets go at the instant the dart leaves the gun. **Show that the dart will *always* hit the monkey**, provided that the dart reaches the monkey before he hits the ground and runs away.

Monkey and Hunter Problem

Element	Behavior
Dart	Projectile motion (parabola) under gravity
Monkey	Free fall (dropped from rest) under gravity
Aim	Dart is aimed directly at the monkey's initial position
Timing	Monkey drops at the exact instant the dart is fired
Air resistance	Ignored

The Surprising Claim

Even though the monkey falls, and even though the dart follows a curved path, **the dart always hits the monkey** regardless of the dart's initial speed (as long as it reaches the monkey before the monkey hits the ground).


y
Trajectory of dart
without gravity
Dart's
fall
Dart's
fall
v_0
Dart's fall
Trajectory of dart
with gravity
α_0
O
x
d
Dart's
fall

Example

Passengers on a carnival ride move at constant speed in a horizontal circle of radius 5.0 m, making a complete circle in 4.0 s. **What is their acceleration?**

$$a_{rad} = \frac{4\pi^2 R}{T^2}$$

$$a_{rad} = \frac{4\pi^2 (5)}{(4)^2}$$

Newton's 1st Laws

An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.

Inertia - the tendency of objects to resist changes in their state of motion.

$$\sum \bar{F} = 0$$

A hockey puck sliding on frictionless ice continues moving indefinitely until it hits the boards.

Newton's 2nd Laws

The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

Net force determines how much an object accelerates.

$$\sum \bar{F} = m\bar{a}$$

ushing a empty shopping cart (small mass) vs. a full shopping cart (large mass) with the same force - the empty cart accelerates more.

Newton's 3rd Laws

For every action, there is an equal and opposite reaction.

Forces always occur in pairs between two interacting objects.

$$\bar{F}_{12} = -\bar{F}_{21}$$

When you push against a wall, the wall pushes back on you with equal force.

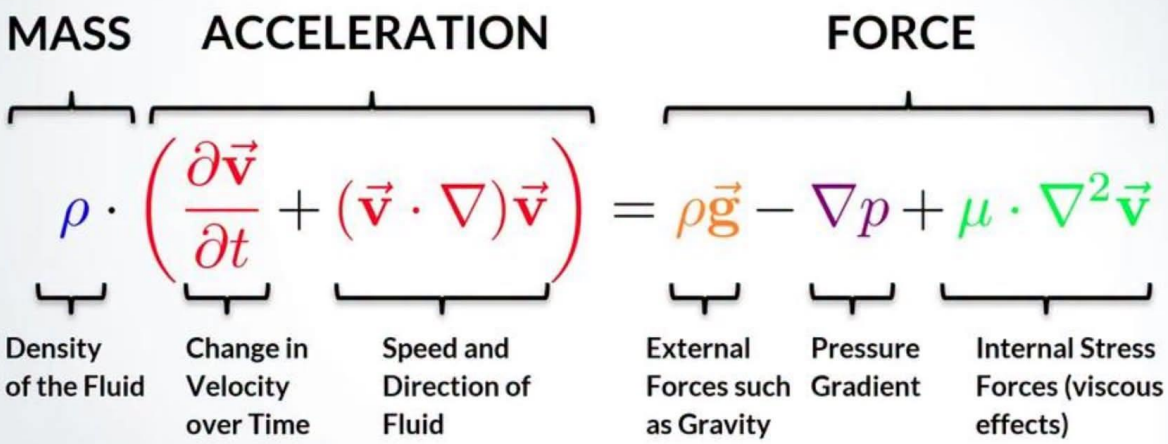
Newton’s Laws of Motion

Law	Physical Quantity	Real-World Application
First Law	Inertia	Seat belts in cars
Second Law	Force, Mass, Acceleration	Rocket propulsion
Third Law	Action-Reaction Pairs	Walking, swimming

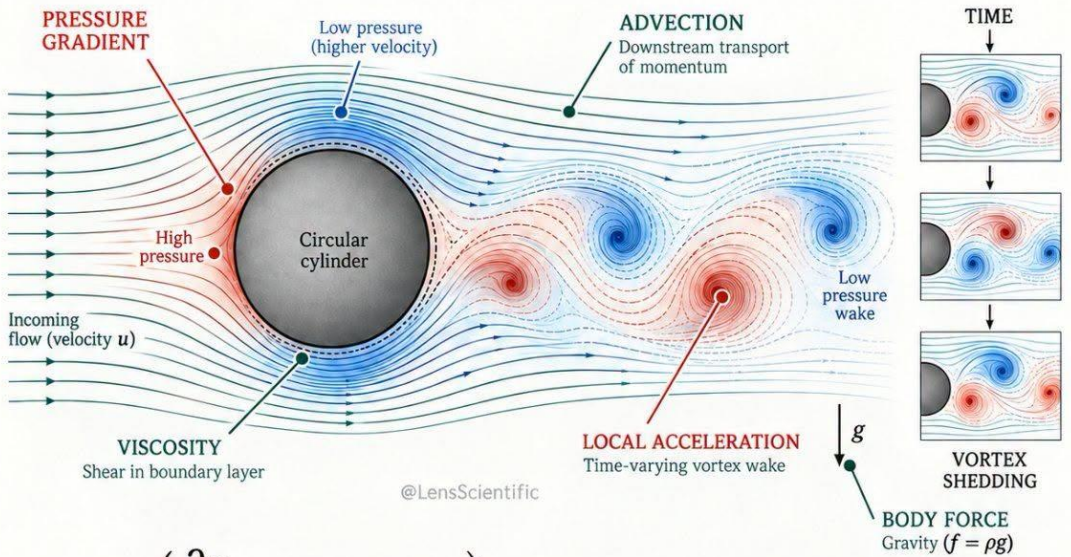
Newton’s Laws of Motion (Limitation)

Limitation	Domain of Failure	Correct Theory
High Speeds (≈ speed of light)	Relativistic Domain	**Special Relativity**
Very Small Sizes (atoms, particles)	Quantum Domain	**Quantum Mechanics**
Gravitational Fields (strong fields, cosmology)	Cosmological Scale	**General Relativity**
Accelerating Reference Frames	Non-Inertial Frames	Requires **Fictitious Forces**
Origin of Forces	Fundamental Physics	**Standard Model**, **GR**

The Navier-Stokes Equations



NAVIER-STOKES EQUATIONS



$$\rho\underbrace{\left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\cdot\nabla)\mathbf{u}\right)}_{\text{INERTIA PER UNIT VOLUME}} = -\nabla p + \mu\nabla^2\mathbf{u} + \mathbf{f}$$

$$\underbrace{\nabla\cdot\mathbf{u} = 0}_{\text{incompressible flow}}$$

$\mathbf{u}$ velocity ρ density p pressure μ dynamic viscosity f body-force density

The equation tracks how momentum moves through a fluid.

It governs water, air, blood flow, weather, and turbulence.

Example

An Aston Martin V12 Vantage sports car has a "lateral acceleration" of 0.97g = (0.97)(9.8 m/s^2) = 9.5 m/s^2. This is the maximum centripetal acceleration the car can sustain without skidding out of a curved path. If the car is traveling at a constant 40 m/s (about 89 mi/h, or 144 km/h) on level ground, what is the radius R of the tightest un-banked curve it can negotiate?

$$a_{rad} = \frac{v^2}{R}$$

$$9.5 = \frac{(40)^2}{R}$$

$$R = \frac{(40)^2}{9.5} = 170m$$

A Comprehensive Problem-Solving Framework

IDENTIFY

Purpose: Understand what the problem is asking and what physical principles apply.

Key Actions:

Read the problem carefully and determine the **target variable(s)** - what you need to find

Identify the **relevant physical concepts** (Newton's laws, conservation of energy, kinematics, etc.)

List all **known quantities** with their values and units

Recognize any **hidden information** or assumptions

Determine the **type of problem** (equilibrium, motion, energy transformation, etc.)

Example Questions to Ask:

What is being asked?

What physical principles govern this situation?

What information is explicitly given?

What can I infer from the context?

A Comprehensive Problem-Solving Framework

SET UP

Purpose: Develop a plan for solving the problem.

Key Actions:

Draw a **diagram** illustrating the situation

Choose an appropriate **coordinate system**

List the **fundamental equations** that relate the known and unknown quantities

Decide on the **solution strategy** (which equations to use in what order)

Define symbols clearly and consistently

Example Questions to Ask:

How can I visualize this problem?

What coordinate system will make the math simplest?

Which equations connect what I know to what I need to find?

What is my step-by-step plan?

A Comprehensive Problem-Solving Framework

EXECUTE

Purpose: Carry out the solution plan mathematically.

Key Actions:

Solve equations **algebraically first** when possible

Substitute numerical values **only after** obtaining algebraic solutions

Show all steps clearly and logically

Include units in all calculations

Carry out calculations carefully

Box your final answer

Example Questions to Ask:

Am I following my planned approach?

Are my mathematical steps correct?

Are units consistent throughout?

Is the algebra sound?

A Comprehensive Problem-Solving Framework

EVALUATE

Purpose: Check whether the solution makes physical sense.

Key Actions:

Check that **units** are correct for the final answer

Assess whether the **magnitude** is reasonable

Consider **limiting cases** (what if a parameter approaches zero or infinity?)

Verify the answer using an **alternative method** when possible

Check the **sign/direction** of vector quantities

Consider whether the answer **makes physical sense**

Example Questions to Ask:

Do the units make sense?

Is the answer numerically reasonable?

Does the behavior match what I expect physically?

Can I verify this result another way?

Mechanics

Type of Mechanics	Key Focus	Sub-Types	Real-World Example
Classical	Rigid Bodies	Statics, Dynamics	Forces in a truss bridge
Continuum	Deformable Bodies	Solid Mechanics, Fluid Mechanics	Airflow over a car (aerodynamics)
Relativistic	High-Speed (≈ c)	Special & General Relativity	GPS satellite corrections
Quantum	Atomic Particles	Quantum Theory, Field Theory	Behavior of electrons in a semiconductor